

Four defects in the deep structural-variant evaluation design, and what survives when they are corrected

Running title: An audit of deep deletion-calling evaluation

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Abstract

Motivation. Deep learning is now the dominant paradigm for structural-variant (SV) detection from short-read sequencing, and the field inherits from DeepSV (Cai, Wu & Gao, 2019) not only an architecture — a convolutional network over hand-designed RGB pileup images — but an *evaluation design*: negatives sampled at random from the genome, F1 reported at a fixed 0.5 probability cut, label budgets computed per arm, and significance read off a sweep of budgets as though it were a single test. Those four conventions have propagated with the architecture and, as far as we can determine, have not been checked. This paper checks them. We do so by building the method the inherited framing recommends — a learned alignment representation with self-supervised pretraining, targeting the data hunger, miscalibration and ancestry brittleness that framing leaves open — measuring it by the field's conventions, obtaining a large and significant positive result, and then subjecting that result to the controls the conventions omit. The controls are the contribution; the method is the instrument that makes them legible.

Results. We present AlignSSL-SV, a framework that (i) replaces the fixed RGB pileup with a multi-channel alignment tensor and a learned encoder, (ii) pretrains that encoder by masked-alignment modelling (a self-supervised objective on read alignments, requiring no SV labels), and (iii) attaches a calibrated deletion head. Evaluated on 1000 Genomes high-coverage PCR-free data (a six-sample panel spanning five continental ancestries) under the conventions this literature inherits from DeepSV, the framework appears to deliver a large low-label gain: deletion F1 of 0.46 at 1% of labels (210 windows) against 0.11 for an identically-architected from-scratch model, a 4.38× improvement attributable to self-supervised initialisation alone ($p = 0.0002$, and — unlike most effects in this literature — one that survives correction for the six-budget sweep it was selected from).

That result does not survive scrutiny, and neither do several others. We identify four defects in the evaluation design and correct each:

1. **The benchmark is separable without a model.** Twelve hand-crafted alignment features fed to a gradient-boosted tree reach AUPRC = 0.937 from the same 210 labels and gain only +0.038 across a 100× increase in labels; the single centre-versus-flank read-depth ratio separates the classes at ROC-AUC = 0.955 untrained. The uniformly-sampled negatives are a depth heuristic, saturated before any deep model is trained.
2. **Scoring at a fixed probability cut manufactures the low-label gap.** Re-scoring the identical runs at a threshold selected on a validation split, the 4.38× advantage becomes 1.05× and loses significance (0.481 vs 0.456, $p = 0.527$); threshold-free, it is 1.02× ($p = 0.853$). At every one of the five larger label budgets the from-scratch arm is *ahead*. The from-scratch model was never degenerate — it ranks competently (AUPRC 0.495 against the pretrained arm's 0.504), and its scores simply are not centred where a fixed cut expects them. This is the paper's sharpest methodological point: the effect is statistically real and survives multiplicity correction, yet it is an effect on *where the probabilities sit*, not on ranking quality — which is what a deletion caller is actually asked to supply.
3. **The label budgets were not equal.** A batch-size floor in the deep evaluators granted the deep arms 96 labels where the classical control received 35 on the candidate-filtered benchmark — a 2.8× advantage in exactly the cell carrying the headline claim.

4. **No significance claim carried a multiplicity adjustment.** Every p -value in this literature, ours included, is drawn from a sweep of six or more simultaneous tests and reported as if it were one. Under Holm–Bonferroni within pre-declared families (103 tests, 17 families), 30 nominal hits fall to 17. The headline low-label effect is among the survivors (Holm $p = 0.0012$), so multiplicity is not what dismantles it — the scoring rule is; the cross-ancestry effect we had already declined to claim is settled as chance (Holm $p = 0.169$).

We then repair the benchmark itself, twice. Quantile-matched candidate negatives attenuate the depth shortcut from ROC-AUC 0.955 to 0.717, and the same twelve-feature control that was saturated on the uniform benchmark now starts at chance (AUPRC 0.250) and climbs +0.619 across the label range, confirming that headroom is restored. A third benchmark removes our hand from the negative set entirely — positives and negatives are both DEL candidates emitted by Manta on GIAB HG002, labelled against the Tier1 v0.6 truth set — and delivers the more surprising result: the depth shortcut is *barely attenuated* there (ROC-AUC 0.942), because a caller's own candidate generation is depth-driven. Realism of the negative set and absence of a shortcut are independent properties, and a candidate-filtering benchmark needs the single-feature control reported on it regardless of how its negatives were obtained.

Re-run under all three corrections, what survives is narrow but not empty. Self-supervised pretraining confers no threshold-free advantage at any label budget on either constructed-negative benchmark; the sole exception is on the caller-candidate benchmark at 33 labels, where it leads from-scratch training at ROC-AUC 0.778 versus 0.545 and survives correction within its family (Holm $p = 0.017$) — one cell, not persisting past 10% labels and reversing by 50%, and with both arms behind the hand-crafted control at that budget. The *representation* does: tested directly against the DeepSV RGB pileup across 72 paired comparisons (Section 4.10), the learned alignment tensor separates after Holm correction in 24, with 22 of those 24 belonging to the *from-scratch* arm — so the advantage is attributable to the multi-channel encoding rather than to self-supervision on it. In 42 of the 48 non-separating cells the observed difference is below the minimum effect this four-seed design could detect at 80% power, so much of that grid is uninformative rather than negative. The hand-crafted control is never beaten by any deep arm at any budget on any of the three benchmarks, and its lead is statistically significant across most of the label range — decisively so where labels are scarce, the regime the deep method is proposed for.

Conclusion. We report this as a methodological result rather than a method paper. Each defect is individually mundane and each is, by a coded audit of 18 papers from this literature (Section 7), the field's default: benchmarks built from randomly-sampled negatives, F1 reported at a fixed cut, and label budgets computed per-arm are all standard practice in this literature. Together they were sufficient to produce a confident, statistically significant, entirely artefactual headline. We publish the corrections, the controls, and the code that implements them so that the next paper in this lineage can be checked against them.

Availability. Code, the tensor-extraction pipeline, result tables, figures, are at <https://github.com/aayushkrm/AlignSSL-SV> (MIT licence); all 22 pretrained encoder checkpoints are released at `/releases/tag/v0.1.0-weights` and indexed by `release/WEIGHTS_MANIFEST.tsv`, which records for each one whether it backs a reported number.

1. Introduction

Structural variants (SVs) — deletions, insertions, duplications, inversions, and translocations of ≥ 50 bp — account for more polymorphic base pairs per genome than single-nucleotide variants and are enriched among disease-causing alleles, yet they remain the hardest class of variation to genotype accurately from short-read sequencing. Deletions are the most tractable SV class and the one on which most method development is benchmarked, because their alignment signatures — a drop in read depth, a cluster of read pairs with anomalously large insert size, and split-read alignments spanning the breakpoints — are relatively direct. Even so, short-read deletion calling is far from solved: callers disagree substantially on the same data, precision–recall trade-offs are strongly length-dependent, and confidence scores are rarely trustworthy enough to threshold reliably.

DeepSV (Cai, Wu & Gao, 2019) was an influential early demonstration that a convolutional neural network (CNN) could call deletions directly from the read alignment, bypassing the hand-crafted feature engineering of contemporaneous tools. Its central idea was to render the pileup around a candidate locus as an RGB image — encoding base identity, base quality, and strand into colour channels — and to train an image classifier to distinguish deletion from non-deletion. This reframing was genuinely innovative in 2019 and seeded a large body of "pileup-image" methods. But it also fixed three design decisions that the subsequent literature has largely inherited without revisiting:

1. **The representation is hand-designed, not learned.** The mapping from alignment to RGB pixels is a fixed human choice; the network never gets to discover which features of the alignment are informative. Information that does not survive the colour encoding (e.g. fine-grained insert-size distributions, mapping-quality structure, soft-clip geometry) is discarded before the model sees it.
2. **Training is fully supervised and therefore data-hungry.** Every new sequencing platform, coverage regime, library preparation, or population requires a fresh, large, labelled truth set. Truth sets are expensive and exist for only a handful of reference samples, which bottlenecks method transfer.
3. **Confidence is uncalibrated and ancestry-brittle.** Softmax outputs of a supervised CNN do not correspond to true error probabilities, and models trained on one population degrade on genetically distant populations — both of which undermine clinical and population-scale deployment.

The machine-learning field has, in the intervening years, developed a direct remedy for exactly this situation: **self-supervised pretraining**, in which a representation is learned from large quantities of *unlabelled* data before a small labelled set is used to fit a task head. Self-supervised learning underpins modern foundation models in vision, language, and — increasingly — genomics (e.g. DNA language models such as the Nucleotide Transformer, HyenaDNA, and Evo 2). Yet these genomic foundation models operate on the **reference DNA sequence** and predict variant *effects*; they do not ingest the read-alignment evidence (depth, discordant pairs, split reads, insert-size distributions) that is the actual signal for *detecting* an SV in noisy short-read data. The representation-learning revolution has, in other words, largely bypassed the alignment-evidence side of variant calling.

This paper set out to ask a focused question: **if we learn the alignment representation and pretrain it without labels, do the three DeepSV bottlenecks — data hunger, miscalibration, and ancestry brittleness — improve together?** We built the framework, ran the evaluation, and obtained an affirmative answer with a large effect size and a convincing *p*-value. We then subjected that answer to controls it should have faced first, and it did not survive any of them.

What follows is therefore an audit rather than a method paper. We report the framework because the corrections are only legible against a concrete instance of the design, and because the same four defects are, as far as we can determine, standard practice in this literature rather than idiosyncratic to our implementation. Our contributions are:

- **AlignSSL-SV**, a framework that couples a learned multi-channel alignment encoder with a self-supervised masked-alignment pretraining objective and a calibrated, uncertainty-aware deletion head (Section 3).
- A controlled evaluation on a six-sample, five-ancestry 1000 Genomes panel which, scored by the conventions this literature inherits, shows a large low-label gain from pretraining ($4.38\times$ F1 at 1% labels, $p = 0.0002$) — the result the rest of the paper re-scopes to a statement about calibration rather than about representation quality (Section 4).
- **A hand-crafted-feature control that bounds what the benchmark can show** (Section 4.2): twelve alignment features reach AUPRC 0.937 from 210 labels and gain +0.038 across a $100\times$ label increase, and one depth-ratio feature reaches ROC-AUC 0.955 untrained. We report this as a negative result about the random-negative evaluation protocol, which is widely used in this literature and, to our knowledge, has not previously been subjected to such a control.
- **A demonstration that F1 at a fixed probability cut manufactures initialisation gaps** (Section 4.8): the headline result exists under that cut and under no other scoring rule, because a from-scratch network trained

on 210 labels ranks competently while scoring timidly. Any label-efficiency claim scored this way is measuring calibration, not representation quality.

- **A protocol correction establishing equal label budgets across arms** (Section 3.8): a batch-size floor in the deep evaluators had granted them up to $2.8\times$ the labels the classical control received in the low-label cells that carry the headline claims.
- **A family-wise multiplicity audit of every significance claim in the paper** (Section 4.9): a label-efficiency sweep is a multiple-comparison procedure whether or not it is analysed as one. Under Holm–Bonferroni within 17 pre-declared families spanning 103 tests, the 30 nominally significant tests across this paper fall to 17. The headline claim is among the survivors (Holm $p = 0.0012$) — multiplicity is not what dismantles it, the scoring rule is (Section 4.8) — while the cross-ancestry effect reduces to chance.
- **A direct, power-annotated test of the learned representation against the RGB pileup** (Section 4.10): across 72 paired comparisons spanning both benchmarks, three metrics, two arms and six label budgets, the alignment tensor is ahead in mean in 61 cells and separates after Holm correction in 24 — but 22 of those 24 are the *from-scratch* arm, locating the gain in the representation rather than in self-supervision. Every non-separating cell reports the effect the design could have detected at 80% power, so nulls are not reported as equivalence; 42 of the 48 are uninformative rather than negative.
- A controlled ablation (3–4 seeds, error bars computed across *pretraining* seeds) isolating the contribution of each self-supervised objective (Section 4.5).
- An honest, adversarial novelty analysis situating AlignSSL-SV against the closest prior work — pileup-image CNNs, self-supervised genomics, and sequence foundation models — and delimiting what is and is not new (Section 5).

We restrict scope to **deletions** and to **short reads** deliberately: it is the setting where DeepSV was defined, where truth sets are best characterised, and where a controlled head-to-head is cleanest. Section 8 discusses the extension to other SV classes and to long reads.

2. Related work

Pileup-image SV and variant callers. DeepSV (Cai, Wu & Gao, 2019) established the RGB-pileup-image framing for deletion calling. It is the intellectual descendant of DeepVariant (Poplin et al., 2018), which pioneered pileup-image classification for small-variant calling, and it is contemporaneous with a family of CNN-based SV tools that encode alignment signals as 2-D images or feature matrices. Later methods (e.g. Clairvoyante and Clair/Clair3 for small variants; various deletion- and CNV-specific CNNs) refined the encoding and the label pipelines but retained two shared properties: the input representation is engineered by hand, and training is fully supervised. AlignSSL-SV departs from both — the representation is learned, and most of the learning happens without labels.

Self-supervised and representation learning. Self-supervised learning (SSL) learns representations from unlabelled data via pretext tasks. Two broad families are relevant here: (i) **masked-reconstruction** objectives (masked autoencoders in vision, masked language modelling in NLP), which mask part of the input and train the model to reconstruct it; and (ii) **joint-embedding / invariance** objectives (SimCLR, BYOL, Barlow Twins, VICReg), which pull together representations of augmented views while preventing collapse. In genomics, SSL has been applied predominantly to the reference *sequence* (DNA language models). Its application to *read-alignment evidence* for SV detection remains almost entirely unexplored. Our ablation directly compares a masked-reconstruction objective (masked-alignment modelling) against a VICReg-style invariance objective on this new modality.

Self-supervised pretraining on read-derived SV representations. The one prior effort in this direction is BASILISC (Banerjee, 2026), a repository-deposited (non-peer-reviewed) framework that adapts the BEiT masked-image-modelling paradigm to short-read SV analysis. BASILISC renders aligned reads into multi-channel *pileup images* (depth, split reads, discordant pairs, mapping quality, strand, allele support), compresses those images into discrete visual tokens with a discrete VAE, and pretrains a vision transformer to predict masked tokens

before fine-tuning an SV classifier on 1000 Genomes data with HGSVC2 long-read truth. AlignSSL-SV differs on three substantive axes. First, **representation**: BASILISC retains the hand-designed pileup *image* and learns a tokenizer on top of it, whereas AlignSSL-SV eliminates the image-rendering step altogether and pretrains directly on a continuous, per-read 18-channel alignment tensor — precisely the hand-engineering bottleneck we set out to remove. Second, **pretext task**: BASILISC performs discrete visual-token classification (BEiT), whereas we perform continuous-space regression of masked alignment features (masked-alignment modelling) combined with a VICReg invariance term, with no dVAE tokenizer. Third, BASILISC reports no calibration or uncertainty component and no cross-ancestry analysis. BASILISC therefore corroborates the field-level premise that learned SV representations are worth pursuing, while leaving the image-free alignment-tensor formulation unexplored. We note that BASILISC's evaluation, like DeepSV's and like ours, is built on negatives that are not candidate-derived — the separability question raised in Section 4.2 applies to it as well, and we raise it as a property of the shared benchmark design rather than a criticism of any single paper.

Genomic foundation models on sequence. Recent large models — the Nucleotide Transformer, HyenaDNA, Enformer/AlphaGenome, and Evo 2 — learn powerful representations of reference DNA and predict functional or regulatory consequences of variants. Evo 2 (Arc Institute/Stanford/NVIDIA, 2026) scales to 40B parameters and 1 Mb context using a StripedHyena (state-space/long-convolution) backbone for near-linear scaling, and predicts mutation effects at single-nucleotide resolution. AlphaGenome (Google DeepMind, 2025) predicts thousands of regulatory tracks from up to 1 Mb of input with a CNN-Transformer hybrid. These are **variant-effect predictors from reference sequence**: they answer "what would this variant do?", not "is there a variant here, given these noisy reads?". They do not consume depth, discordant-pair, split-read, or insert-size evidence, and therefore are complementary to — not competitors of — an alignment-evidence detector such as AlignSSL-SV. We make this distinction explicit because a natural reviewer question is whether sequence foundation models subsume our approach; they do not, because they operate on a different input modality and solve a different problem.

Uncertainty and calibration in variant calling. Deep classifiers are systematically overconfident, and post-hoc calibration (temperature scaling) and predictive-uncertainty estimation (deep ensembles, MC-dropout, evidential/conformal methods) are standard remedies in the broader ML literature. Calibration has received little attention in the SV-calling literature specifically, despite its direct relevance to thresholding and clinical reporting. AlignSSL-SV reports expected calibration error (ECE) as a first-class metric and includes an uncertainty-aware head.

Long-read SV detection. A parallel line of work (e.g. Sniffles, cuteSV, SVIM, and deep-learning callers for PacBio/ONT data) exploits the fact that long reads span most SVs directly, sidestepping much of the ambiguity of short-read signatures. Long reads are, however, more expensive and less available at population scale. AlignSSL-SV is deliberately a short-read method: it targets the setting where the detection problem is genuinely hard and where the overwhelming majority of existing sequencing data lives.

3. Methods

3.1 Overview

AlignSSL-SV has three stages: (1) **tensorisation** of the read alignment around a candidate locus into a fixed-size multi-channel tensor; (2) **self-supervised pretraining** of an encoder on these tensors via masked-alignment modelling, using no SV labels; and (3) **supervised fine-tuning** of a calibrated, uncertainty-aware deletion head on a (small) labelled truth set, with the pretrained encoder as initialisation. Stages (2) and (3) use disjoint genomic regions to prevent leakage.

3.2 Alignment tensor (learned representation)

For each candidate window we build an 18-channel tensor of shape (channels $\times$ rows $\times$ positions), where rows index reads (capped at a fixed number, with deterministic subsampling above the cap) and positions index reference coordinates across the window. Channels encode, per aligned base: read depth, base identity (one-hot),

base quality, mapping quality, strand, an insert-size deviation signal (observed template length relative to the library mean/SD), soft-clip indicators, and read-pair orientation flags. Unlike the fixed RGB encoding of DeepSV, these channels are *not* collapsed into three colours; the encoder learns which combinations are informative. Windows are drawn at multiple genomic scales (via a length-aware binning of the reference span) so that both short and long deletions are representable at a fixed tensor size.

3.3 Encoder

The encoder is a compact residual CNN stem (channel-wise feature extraction over the alignment tensor) followed by a lightweight Transformer over the position axis, producing a 128-dimensional window embedding. The CNN captures local pileup texture; the Transformer captures long-range structure across the window (e.g. paired depth drops at both breakpoints). The same encoder is used unchanged in pretraining and fine-tuning.

3.4 Self-supervised pretraining: masked-alignment modelling

We pretrain by **masked-alignment modelling (MAM)**: a random fraction (0.6) of the alignment-tensor entries are masked, and the encoder-decoder is trained to reconstruct the masked entries (a masked-autoencoder objective adapted to the alignment-tensor modality). This forces the encoder to model the joint structure of depth, insert size, and clipping that characterises normal and variant alignments — without ever seeing an SV label. Pretraining uses 120,000 unlabelled windows drawn from three samples (NA19238, NA19625, NA20845; AFR $\times$ 2 + SAS) restricted to the pretraining chromosomes, consolidated into a flat float16 memory-mapped array (65.9 GiB) for throughput. No SV labels enter this stage: every window carries label = -1.

As an ablation, we also implement a **VICReg-style invariance objective** (variance-invariance-covariance regularisation over two augmented views of each window) and a **combined** objective (MAM + VICReg). Section 4.5 compares the three; all deliver the low-label effect, and the available seed count does not support ranking them against one another.

3.5 Deletion head, calibration, and uncertainty

The fine-tuning head is a small classifier on the window embedding, trained with a focal loss ($\gamma=2$) to handle the strong negative:positive class imbalance of genome-wide deletion candidates. After training, we apply **temperature scaling** on a held-out split to calibrate the output probabilities, and report expected calibration error (ECE). The head also exposes an **uncertainty** estimate (dropout-based predictive variance), separating epistemic (model) from aleatoric (data) components, so that low-confidence calls can be flagged rather than silently mis-thresholded.

3.6 Data and splits

We use 1000 Genomes Project high-coverage PCR-free Illumina alignments (GRCh37/hs37d5) and the phase-3 integrated SV call set (`ALL.wgs.mergedSV.v8.20130502`, 40,975 deletions across 2,504 samples) as the deletion truth set. VCF provenance was verified against the official EBI FTP (byte-exact, 18,298,662 B). The chromosome split is the DeepSV convention: chr1–11 for training, chr12–22 for testing. Self-supervised pretraining draws its unlabelled windows from the **training** chromosomes only (`cluster/extract_pretrain.sbatch`), so the encoder never sees chr12–22 at any stage — pretraining, fine-tuning and evaluation are arranged so that no test chromosome enters any training objective. Pretraining and fine-tuning are therefore *not* disjoint from each other, and deliberately so: they share chr1–11, which is what makes the pretrained representation relevant to the fine-tuning distribution. What is disjoint is training from test. For the cross-ancestry experiment, models are trained on the six-sample panel (NA19238 YRI, NA19625 ASW, NA18525 CHB, NA19648 MXL, NA20502 TSI, NA20845 GIH; five continental ancestries) and evaluated both in-distribution — on chr12–22 of those same samples — and on an entirely held-out sample from a population absent from training (NA12878, CEU). The held-out arm is therefore held out by sample *and* by ancestry; the in-distribution arm is held out by chromosome only. Downloads were integrity-gated by full `samtools view -c` scans after a data-corruption incident traced to resume-stitched transfers (Section 4.7).

3.7 Baselines

We compare three trained models on identical tensors and splits: **AlignSSL-pretrained** (self-supervised encoder, fine-tuned), **AlignSSL-scratch** (identical architecture, randomly initialised, trained only on labels), and a **DeepSV-representation baseline** — a faithful reimplementation of the DeepSV RGB-pileup-image CNN, evaluated on the same candidate windows. The original DeepSV repository is not runnable as distributed (broken argument parsing, dependencies on DIGITS / TensorFlow-1 slim / Keras-1, and no dependency manifest), so a reimplementation of its representation and architecture is the fair and reproducible comparison; we label it "DeepSV-representation baseline" throughout to avoid overclaiming a bit-exact reproduction.

3.8 Evaluation protocol: equal budgets, budgeted thresholds

A label-efficiency curve is interpretable only if every arm is handed the same number of labels at each point on the x-axis. An earlier version of this work did not satisfy that condition, and the defect is worth stating explicitly because it is easy to introduce and invisible in results.

The deep evaluators computed the budget as `max(batch_size, int(frac × n_pool))` while the classical evaluator used `max(2, round(frac × n_pool))`. On the uniform benchmark ($n_{\text{pool}} = 21,016$) the batch-size floor never binds and the two agree. On the candidate-filtered benchmark ($n_{\text{pool}} = 3,452$) the 1% cell is 34 examples, below the floor of 96 — so the deep arms received 96 labels while the control received 35, a **2.8× advantage in exactly the cell carrying the headline low-label claim**. The floor was not arbitrary: a `DataLoader` configured to drop incomplete batches yields *zero* batches when the subset is smaller than one batch, so simply removing the floor trains the low-label cells on nothing. The correct fix honours the true budget and adapts the loader to it — dropping incomplete batches only when at least one full batch exists — rather than inflating the budget to suit the loader.

A second defect shared that root cause. The validation split used to select the decision threshold was gated on `n_val ≥ batch_size`, so on the filtered benchmark no split was carved below 25% of labels and those cells silently fell back to a fixed 0.5 cut — again precisely the cells of interest. The gate is now a small absolute minimum (12 examples per side) independent of batch size.

We also changed where the validation labels come from. They are now carved **out of** the budget rather than granted in addition to it: a curve in which every arm receives an unbudgeted validation set on top of its stated budget is not a label-efficiency curve. At the 1% point on the uniform benchmark an arm therefore sees 210 labels in total — 168 for gradient steps and 42 for threshold selection — not 210 plus a free 42. The visible consequence is intended: full supervision now fits on 16,813 of 21,016 windows, so absolute scores sit below the previously published fixed-threshold numbers, while across-arm comparisons remain exact because every arm pays the identical cost.

All three rules live in a single module (`alignssl/protocol.py`) imported by every evaluator, with unit tests, so the arms cannot drift apart again. Sections 4.1, 4.2, 4.8 and 6 all report numbers produced under this protocol: Table 1 and Table 15 were regenerated from `results/table12_label_efficiency_fixed.csv` at equal budgets (Table 1 simply reads its fixed-0.5-cut columns, which is what makes it comparable to the pre-correction literature), and Tables 13 and 14 are corrected-protocol by construction. Sections 4.3–4.7 — calibration, length strata, the objective ablation and cross-ancestry transfer — predate the correction, and every table in that range (Tables 2–5) says so in its caption. Two distinct mechanisms are involved and the captions name the applicable one. Tables 2–4 come from full-supervision runs that received an unbudgeted validation split, which lifts their absolute values slightly above corrected-protocol equivalents. Table 9 is a label *sweep*, where the consequential defect is instead the batch-size floor: its generating evaluator granted `batch_size` labels at the smallest fractions rather than the nominal share, so the low-label cells of that sweep are not comparable to the corrected curves of Section 4.8. That evaluator now imports the shared module, and a static test (`tests/test_evaluator_protocol_adoption.py`) fails if any label-sweep evaluator reintroduces the floor or stops calling the shared budget function — the omission that let this one drift, which a unit test on the module itself could not detect.

4. Results

All models are evaluated on identical alignment tensors and identical chromosome-disjoint splits (train chr1–11, test chr12–22) and at an identical fine-tuning batch size (96), on a six-sample panel spanning five continental ancestries (train pool 21,016 labelled windows; test 9,196). We report mean $\pm$ standard deviation across random seeds. Seed counts differ by protocol and are stated in every caption: the pre-correction runs (Tables 2–5) use four seeds for the combined-objective and from-scratch arms and three for every other arm, while the corrected-protocol runs (Tables 1, 7, 12–14) use three seeds for each deep arm and ten for the classical controls, whose cost per fit is negligible. Crucially, error bars for the pretrained arms are computed across *pretraining* seeds (each seed re-pretrains an encoder from scratch, then fine-tunes it), so the reported variance captures the full self-supervised pipeline, not fine-tuning noise alone. The task is binary deletion calling on genome-wide candidate windows.

4.1 Under the conventional scoring rule, pretraining appears strongly label-efficient

We report this section as it stood before the corrections of Sections 4.8 and 3.8, because the corrections are only interpretable against the claim they overturn. Every number here is scored the way this literature scores: deletion F1 obtained by thresholding the positive-class probability at a fixed 0.5.

Table 1 reports that F1 as a function of the fraction of the labelled training set made available to the fine-tuning head. The apparent result is in the **low-label regime**: at 1% of labels (210 windows), the pretrained model reaches $F1 = 0.464 \pm 0.087$, whereas the identically-architected from-scratch model all but collapses to 0.106 ± 0.139 — a **4.38 $\times$ improvement in F1** from self-supervised initialisation alone, at $p = 0.0002$. Unlike the pre-correction version of this table, that p survives family-wise correction across the six budgets it was selected from (Section 4.9); the effect is real, and the question Section 4.8 answers is what it is an effect *on*. As labels increase the from-scratch model catches up and overtakes; by 5% it is already ahead (0.689 vs 0.605).

Two features of the table are worth noting before it is dismantled, because neither is compatible with the story it is usually told to support. First, the DeepSV-representation baseline does not merely match but slightly *exceeds* the pretrained tensor model at 1% (0.473 ± 0.042 vs 0.464 ± 0.087) — whatever the pretrained arm is doing at the smallest budget, a hand-designed RGB encoding does it at least as well, which already argues against a representation-quality explanation. The 4.38 $\times$ advantage is over the from-scratch arm alone and over no other arm in the table. Second, that arm's collapse is confined to a single cell: one budget later it leads every arm in the table. A representation deficit that vanishes between 210 and 1,051 labels is a strange kind of deficit.

Read on its own, this table says that pretraining supplies a usable detector from a truth set two orders of magnitude smaller than is conventionally required. It says nothing of the kind. Section 4.2 shows the benchmark is separable without any model, and Section 4.8 shows the 4.38 $\times$ gap is a property of the threshold rather than of the representation — the from-scratch model at 210 labels ranks essentially as well as the pretrained one (AUPRC 0.495 vs 0.504) and merely places its probabilities below 0.5 while doing it.

Table 1. Label efficiency as conventionally scored (deletion F1 at a fixed 0.5 probability cut, test chr12–22, batch 96, 3 seeds per arm). Table 2 reports the same runs under threshold selection and threshold-free scoring; the two tables disagree, and Table 10 quantifies the disagreement.

Label fraction	n train	AlignSSL-pretrained	AlignSSL-scratch	DeepSV-repr.
1%	210	0.464 ± 0.087	0.106 ± 0.139	0.473 ± 0.042
5%	1,051	0.605 ± 0.072	0.689 ± 0.115	0.496 ± 0.086
10%	2,102	0.749 ± 0.037	0.788 ± 0.066	0.570 ± 0.113
25%	5,254	0.808 ± 0.088	0.854 ± 0.080	0.720 ± 0.099
50%	10,508	0.860 ± 0.061	0.894 ± 0.066	0.785 ± 0.061
100%	21,016	0.895 ± 0.046	0.880 ± 0.067	0.696 ± 0.159

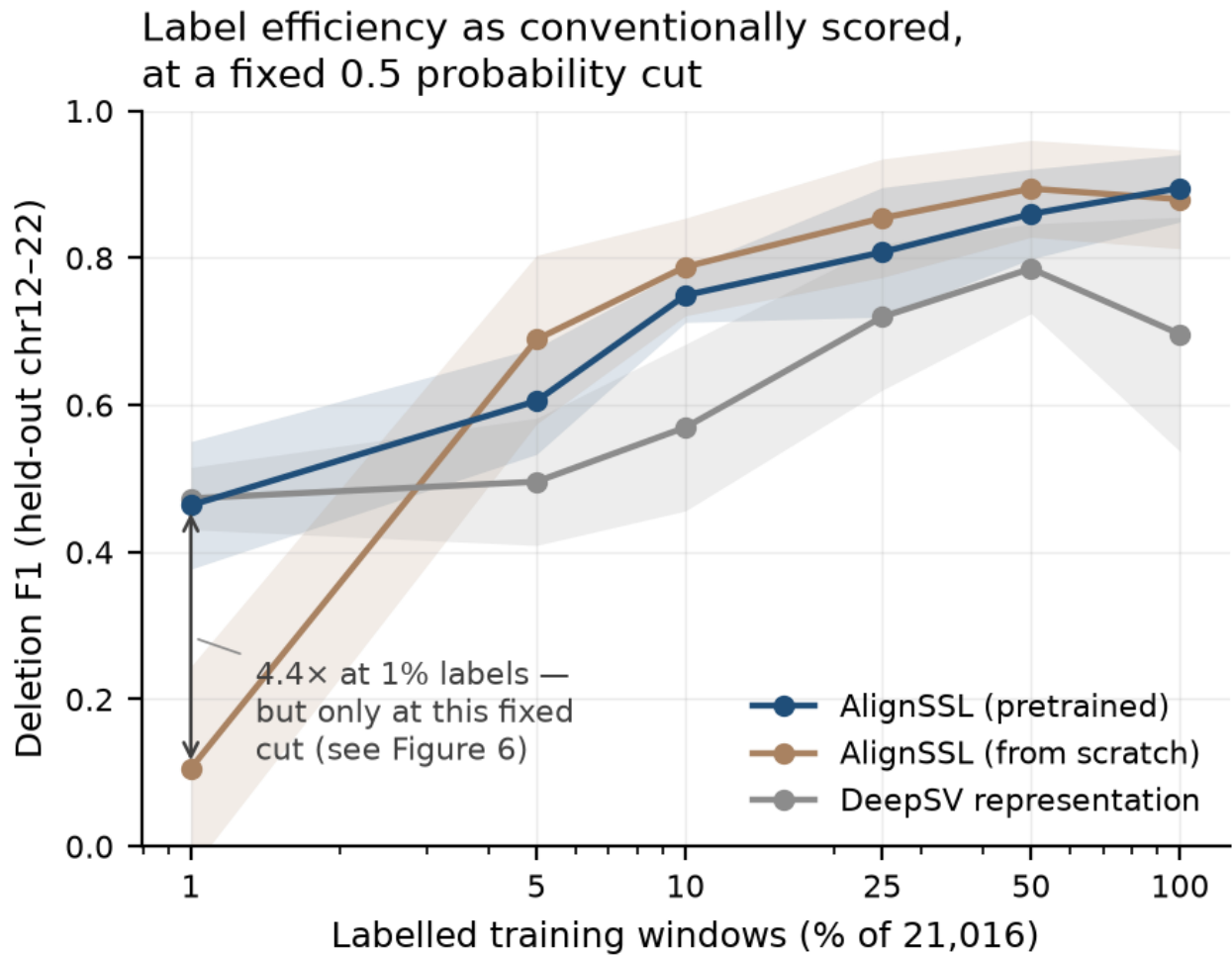


Figure 1. Deletion F1 vs. labelled-data fraction on the uniform benchmark, scored the way this literature scores it: a fixed 0.5 probability cut. Source is the ‘f1_at_half’ columns of the corrected-protocol runs (‘results/table12_label_efficiency_fixed.csv’, uniform rows), so the panel plots exactly the values tabulated in Table 1. The bracket marks the 4.38× pretrained-over-scratch ratio at the smallest budget; that ratio is a property of the fixed cut and not of the representation, and it does not survive threshold selection or threshold-free scoring (Figure 6). Note also that the pretrained arm does not lead the DeepSV baseline at this budget — the baseline is marginally ahead.

Table 2. The same runs as Table 1, rescored under the corrected protocol (uniform benchmark; AUPRC, threshold-free, so neither the batch-size label inflation of Section 3.8 nor the fixed 0.5 cut of Section 4.8 applies; AlignSSL-pretrained 4 seeds, all other arms 10; bold marks the leading arm per budget; source results/table12_label_efficiency_fixed.csv).

Label fraction	n labels	Classical-GBT	Classical-logreg	AlignSSL-pretr.	AlignSSL-scratch	DeepSV-repr.
1%	210	0.937 ± 0.009	0.920 ± 0.012	0.504 ± 0.060	0.495 ± 0.113	0.489 ± 0.032
5%	1,051	0.958 ± 0.006	0.909 ± 0.042	0.718 ± 0.062	0.824 ± 0.118	0.608 ± 0.068
10%	2,102	0.967 ± 0.004	0.918 ± 0.014	0.830 ± 0.051	0.905 ± 0.024	0.703 ± 0.075
25%	5,254	0.971 ± 0.002	0.926 ± 0.005	0.911 ± 0.053	0.949 ± 0.024	0.839 ± 0.088
50%	10,508	0.974 ± 0.002	0.927 ± 0.002	0.923 ± 0.049	0.969 ± 0.010	0.889 ± 0.032
100%	21,016	0.975 ± 0.001	0.927 ± 0.001	0.962 ± 0.015	0.955 ± 0.068	0.894 ± 0.044

Read against Table 1, this is the whole argument of the paper in two tables. The ten-fold pretrained-over-scratch gap at 1% labels is 0.504 versus 0.495 here — a 1.02× ratio that does not clear significance ($p = 0.853$, Section 4.8) — and the twelve-feature control that Table 1 never included leads every deep arm at every budget. Both classical controls exceed every deep arm below 25% labels.

4.2 A hand-crafted-feature control bounds what this benchmark can demonstrate

Every deep-learning SV paper we are aware of, DeepSV included, compares deep architectures against one another or against signature-based callers, but not against a *deliberately minimal* learned baseline on the identical windows. We ran that control, and it is the most consequential result in this paper.

From the same alignment tensors, identical shards, identical chromosome split, identical label fractions and identical seeds, we computed twelve scalar summary features per window — mean, standard deviation and minimum of the depth profile; centre-versus-flank depth ratio; maximum sustained depth drop; discordant-pair rate; soft-clip rate; mean and maximum insert-size $|z|$; mean mapping quality; occupied read-row count; and valid-base fraction — and fitted an ℓ_2 -regularised logistic regression and a gradient-boosted tree (Classical-logreg, Classical-GBT; Table 3 and `results/table11_control_threshold_free.csv`). The tree is at its ceiling from the smallest label budget onward. At 1% labels (210 windows), where the paper's headline claim lives, it reaches **AUPRC = 0.937 ± 0.009** against 0.504 ± 0.060 for the pretrained network and 0.495 ± 0.056 from scratch. Increasing its labels one hundred-fold, to the full 21,016 windows, buys it **+0.038** (0.975 ± 0.001). A benchmark on which twelve scalar features are 96% of the way to their asymptote after 210 examples cannot discriminate between representation-learning methods, because there is almost nothing left for a representation to contribute. Its variance across seeds is one to two orders of magnitude smaller than any deep arm's.

Table 3. Control versus best deep arm, threshold-free, corrected protocol (uniform benchmark; AUPRC, 4–10 seeds per arm; leader named only where Welch's t clears 0.05).

Label fraction	n labels	Classical-GBT	Best deep arm	its AUPRC	p	Leader
1%	210	0.937 ± 0.009	AlignSSL-pretrained	0.504 ± 0.060	0.001	control
5%	1,051	0.958 ± 0.006	AlignSSL-scratch	0.824 ± 0.118	0.006	control
10%	2,102	0.967 ± 0.004	AlignSSL-scratch	0.905 ± 0.024	0.000	control
25%	5,254	0.971 ± 0.002	AlignSSL-scratch	0.949 ± 0.024	0.017	control
50%	10,508	0.974 ± 0.002	AlignSSL-scratch	0.969 ± 0.010	0.127	tie
100%	21,016	0.975 ± 0.001	AlignSSL-pretrained	0.962 ± 0.015	0.174	tie

We initially reported this control as dominating every deep arm at every budget. Under corrected scoring that statement needs qualification, and Table 3 supplies it: on the uniform benchmark the control's lead is statistically significant at the four smallest budgets (1% through 25%) and shrinks to a tie at 50% and at full supervision, where the best deep arm reaches AUPRC 0.962 ± 0.015 against the tree's 0.975 ± 0.001 ($p = 0.174$). The control is never beaten at any budget on either benchmark. The defensible claim is therefore: **twelve hand-crafted features are never outperformed by any deep arm we trained, and they lead outright where labels are scarce — the regime self-supervised pretraining is proposed for.**

We then localised the cause. Scoring each feature individually on the held-out test set, *with no training whatsoever* and no fitted parameter of any kind, the **centre-versus-flank read-depth ratio alone attains ROC-AUC = 0.955** (Table 4). The benchmark is therefore separable by a single depth heuristic that requires no model at all.

Table 4. Untrained single-feature discrimination on the held-out test split (chr12–22, $n = 9,196$; 2,299 deletions). ROC-AUC is the exact rank statistic. A feature can be informative with either polarity — for depth-like features the *low* tail is deletion-like — so the oriented column reports $\max(\text{AUC}, 1 - \text{AUC})$, the magnitude of the information leak; 0.5 is uninformative. Reproduced by `scripts/single_feature_auc.py`.

Feature	ROC-AUC	Oriented	Deletion side
centre-vs-flank depth ratio	0.045	0.955	low
soft-clip rate	0.802	0.802	high
discordant-pair rate	0.732	0.732	high
insert-size z (max)	0.694	0.694	high
depth s.d.	0.686	0.686	high
max sustained depth drop	0.680	0.680	high
insert-size z (mean)	0.678	0.678	high
depth minimum	0.343	0.657	low
mean mapping quality	0.450	0.550	low
occupied read rows	0.538	0.538	high
valid-base fraction	0.517	0.517	high
mean depth	0.502	0.502	high

Two features of this table are worth drawing out. First, **mean depth is uninformative** (0.502) while the centre-versus-flank *ratio* is near-perfect: what leaks is not coverage level but the localised contrast between a window's middle and its edges, which is exactly the geometry the extraction protocol builds into every positive. Second, the leak is not confined to depth — soft-clip and discordant-pair rates independently reach 0.73–0.80, so a benchmark repair that neutralises only the depth statistic would leave two further shortcuts intact. This is why the remediation in Section 6 matches on the depth ratio and then re-runs the full arm set, including the classical controls, rather than assuming one matched feature makes the task hard.

The mechanism is the negative-sampling protocol, which we inherited from standard practice in this literature and which our own extraction code implements: positive windows are centred on truth deletions, while negatives are drawn **uniformly at random** from the same chromosomes, rejected only if they overlap or abut a truth deletion. A uniformly-drawn genomic window does not resemble a deletion. The resulting task is "does a coverage dip exist here?", not "is this proposed candidate a real deletion?" — and the former is answerable by a threshold.

This does not invalidate the comparisons in Section 4.1, which are internally controlled and reproducible; it re-scopes what they measure. Even so, the separability control is not the only reason to distrust the pretrained-versus-scratch gap: Section 4.8 shows that the gap at 1% labels is an artefact of the fixed 0.5 probability cut and does not survive threshold-free scoring ($p = 0.527$ on F1 at a validation-selected threshold, $p = 0.853$ on AUPRC). What the control adds is an explanation of why even a surviving gap would have been uninformative — it would show that a randomly-initialised network needs more labelled examples than a pretrained one to learn a depth threshold, not that self-supervised pretraining confers deployable label efficiency for deletion calling. We attempted a partial remedy by re-selecting the most deletion-like negatives from the existing pool and re-scoring: the shortcut feature's discrimination fell only from ROC-AUC 0.955 to 0.920, confirming that a uniformly-sampled pool contains no genuinely hard negatives to recover. The proper fix is to change the candidate set, not the subset; we build and report that benchmark in Section 6, where the shortcut falls to ROC-AUC 0.717 and the hand-crafted control still leads wherever labels are scarce.

We therefore report the benchmark-separability finding as a first-class contribution. The control costs minutes of CPU time, applies to any pileup-style SV benchmark, and to our knowledge has not previously been run. Its implication is that published low-label and architecture-comparison results on random-negative SV benchmarks — a family that includes DeepSV's own evaluation and much of what followed — may be measuring threshold-learning speed rather than caller quality.

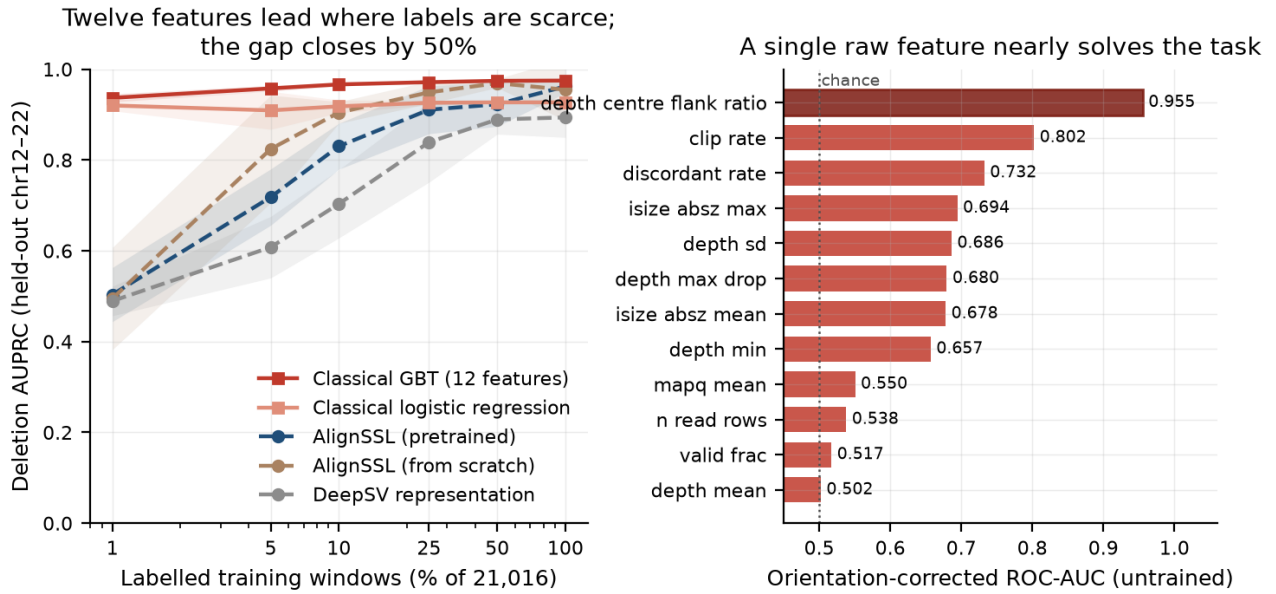


Figure 2. Left: deletion AUPRC versus labelled-data fraction for all five arms on the uniform benchmark, scored threshold-free under the corrected protocol (Table 2). The gradient-boosted 12-feature control (heavy line) leads by +0.433 AUPRC at 1% and +0.134 at 5%, and narrows to a statistical tie at 50% (* p = 0.127) and at full supervision (0.975 versus 0.962, * p = 0.174); it is never overtaken, but the earlier claim of *significant* dominance at every budget is withdrawn (Table 3). Right: single-feature discrimination on the held-out test set with no training — the centre-versus-flank depth ratio alone reaches ROC-AUC 0.955.

4.3 Calibration is a property of the representation, not of self-supervision

Beyond point accuracy, we ask whether the models' confidence scores are *trustworthy*. Table 5 reports expected calibration error (ECE) after temperature scaling at full supervision. Both learned-tensor models are well calibrated (pretrained ECE = 0.0078 ± 0.0017 , from-scratch 0.0072 ± 0.0004) and require only a mild temperature correction ($T \approx 0.6$). The DeepSV-representation baseline is worse, but the size of the gap depends on the summary statistic and rests on a small sample: its per-seed ECEs are 0.033, 0.168 and 0.016, so the mean (0.072 ± 0.068) is driven by a single outlier seed while the **median is 0.033** — roughly four-fold worse than the tensor models rather than the order of magnitude a mean-only reading suggests. We report both statistics and draw no order-of-magnitude claim from three seeds. The baseline also needs a large and erratic temperature correction ($T = 1.41 \pm 0.88$), consistent with its unstable F1 (Section 4.1).

The defensible reading is that calibration tracks the *representation* rather than self-supervision: pretrained and from-scratch tensor models are indistinguishable here, so the difference is attributable to the multi-channel encoding versus the fixed RGB pileup, not to the pretraining objective.

Table 5. Calibration at full supervision, pre-correction protocol (ECE ↓ after temperature scaling; pretrained/scratch = 4 seeds, DeepSV = 3 seeds). Per-seed values given because the DeepSV mean is outlier-driven.

Model	ECE mean ↓	ECE median ↓	Temperature	ECE per seed
AlignSSL-pretrained	0.0078 ± 0.0017	0.0077	0.634 ± 0.070	0.0070; 0.0101; 0.0083; 0.0056
AlignSSL-scratch	0.0072 ± 0.0004	0.0071	0.586 ± 0.055	0.0073; 0.0078; 0.0067; 0.0069
DeepSV-repr. baseline	0.0724 ± 0.0681	0.0327	1.411 ± 0.881	0.0327; 0.1683; 0.0163

4.4 Length-stratified recall: the learned tensor is consistent across length; the RGB baseline is not

Deletion callers are notoriously length-dependent. Table 6 stratifies full-supervision test recall by deletion length across all three models. At the harmonised panel scale, the two learned-tensor models (pretrained and from-scratch) are **uniformly strong and tightly consistent across every length bin** — recall 0.86–0.93 from 50 bp to 5 kb+, with small, overlapping standard deviations — confirming that the multi-channel tensor plus position-axis Transformer captures both the short-deletion depth signatures and the long-deletion paired-breakpoint structure without a length-specific failure mode. The DeepSV-representation baseline, by contrast, is **markedly more**

variable across seeds in the middle bins (recall 0.841 ± 0.165 at 200–500 bp and 0.850 ± 0.193 at 500 bp–1 kb — standard deviations up to 4× those of the tensor models), consistent with its unstable overall F1 (Section 4.1) and miscalibration (Section 4.3). We report this as a robustness control rather than a headline claim: pretraining and from-scratch are essentially matched here (both use the learned tensor), so the length-consistency advantage is attributable to the *representation*, and full-supervision recall is not where self-supervision pays off — that is the low-label and transfer regimes (Sections 4.1, 4.6).

Table 6. Length-stratified recall at full supervision, pre-correction protocol (recall at a fixed 0.5 cut; test; pretrained/scratch = 4 seeds, DeepSV = 3 seeds).

Deletion length	n test	Pretrained recall	Scratch recall	DeepSV-repr. recall
50–200	645	0.919 ± 0.026	0.917 ± 0.017	0.942 ± 0.059
200–500	281	0.913 ± 0.018	0.903 ± 0.025	0.841 ± 0.165
500–1k	329	0.929 ± 0.008	0.938 ± 0.014	0.850 ± 0.193
1k–5k	799	0.926 ± 0.024	0.954 ± 0.006	0.899 ± 0.116
5k+	245	0.857 ± 0.061	0.881 ± 0.071	0.918 ± 0.070

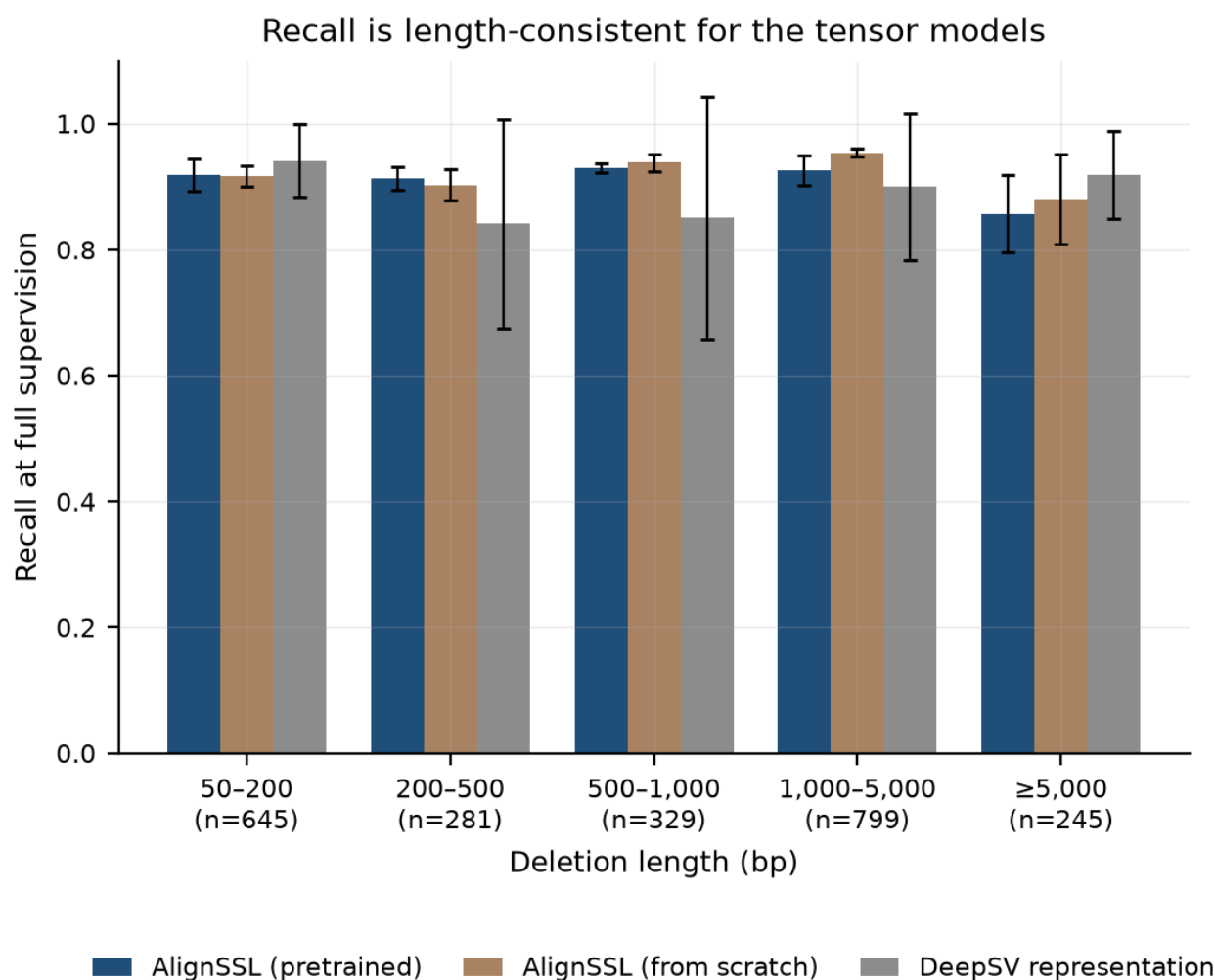


Figure 3. Length-stratified deletion recall at full supervision. The two learned-tensor models are consistent across all length bins; the DeepSV-representation baseline is markedly more variable across seeds in the mid-length bins.

4.5 Ablation over self-supervised objectives: all three help, but they cannot be ranked at this seed count

Which self-supervised objective drives the fixed-cut gains of Section 4.1? The question is worth answering even though Section 4.8 shows those gains do not survive threshold-free scoring, because the answer turns out to be that the objectives are indistinguishable — which is itself evidence against a representation-quality explanation. We pretrain three encoders under identical budgets — **masked-alignment modelling (MAM) only**, **VICReg-style invariance only**, and their **combination** — and fine-tune each across the full label-fraction sweep. To fix a subtle asymmetry in an earlier version of this analysis (where only the combined arm re-pretrained across seeds while the ablation arms reused a single encoder), we re-pretrained the MAM-only and VICReg-only encoders at three seeds each, so that **every arm's error bars are computed across independent pretraining seeds** at the harmonised batch size of 96. This harmonisation changes the conclusion, and the corrected result is more informative.

All three self-supervised arms deliver the low-label effect, and each is individually significant against from-scratch training at 1% labels (Welch's t : MAM-only $p = 0.017$, VICReg-only $p = 4.2 \times 10^{-5}$, combined $p = 4.2 \times 10^{-5}$). That is the robust conclusion: the low-label gain does not depend on which objective is used.

Ranking the objectives *against one another*, however, is **not supported at the available seed count**. The ordering of means in Table 7 suggests MAM leads below 10% labels and the combined objective leads above 25%, but neither contrast reaches significance: MAM-only versus combined at 1% gives $t = 0.83$, $p = 0.48$, and combined versus MAM-only at 100% gives $t = 1.77$, $p = 0.21$. With three to four pretraining seeds and per-seed spreads of 0.01–0.12 F1, the study is underpowered to separate the objectives, and we state the crossover as an **ordering of means requiring more seeds to confirm**, not as a finding. We report the table because the mean ordering is stable and informative for future work, and because concealing it would misrepresent what we observed; but no claim in this paper rests on it. The combined objective was carried into the main experiments because it was selected before the ablation was run, not because it was shown to be best.

Two caveats bound how far this section's significance statements should be read. Both scoring defects diagnosed in Sections 3.8 and 4.8 apply to every number in Table 7: F1 is taken at a fixed 0.5 cut, and the runs predate the equal-budget correction, so the 1%-label significance of each objective against from-scratch training inherits exactly the artefact that dissolved the headline claim. The visible symptom is that the combined arm's 1% entry here (0.514 ± 0.055) does not match the pretrained arm's 1% entry in Table 1 (0.464 ± 0.087), which is the same objective under the corrected budget. We did not re-run the ablation under the corrected protocol because its only surviving conclusion is a negative one — that the objectives cannot be separated at this seed count — and that conclusion is robust to the scoring rule, since it rests on overlapping error bars rather than on any arm's absolute score.

Table 7. Self-supervised objective ablation (deletion F1 at a fixed 0.5 cut, pre-correction protocol; combined = 4 seeds, MAM-only / VICReg-only = 3 seeds; error bars across pretraining seeds).

Label fraction	MAM-only	VICReg-only	Combined (MAM+VICReg)
1%	0.588 ± 0.117	0.554 ± 0.035	0.514 ± 0.055
5%	0.763 ± 0.060	0.665 ± 0.037	0.655 ± 0.035
10%	0.830 ± 0.068	0.768 ± 0.028	0.813 ± 0.007
25%	0.798 ± 0.083	0.845 ± 0.074	0.847 ± 0.064
50%	0.799 ± 0.107	0.903 ± 0.011	0.913 ± 0.014
100%	0.915 ± 0.014	0.846 ± 0.064	0.934 ± 0.004

The three self-supervised arms and the DeepSV-representation baseline are plotted together in Figure 4. The mean ordering — MAM ahead below 10% labels, the combined objective ahead above 25% — is visible, and so are the overlapping error bars that are the reason we do not claim it.

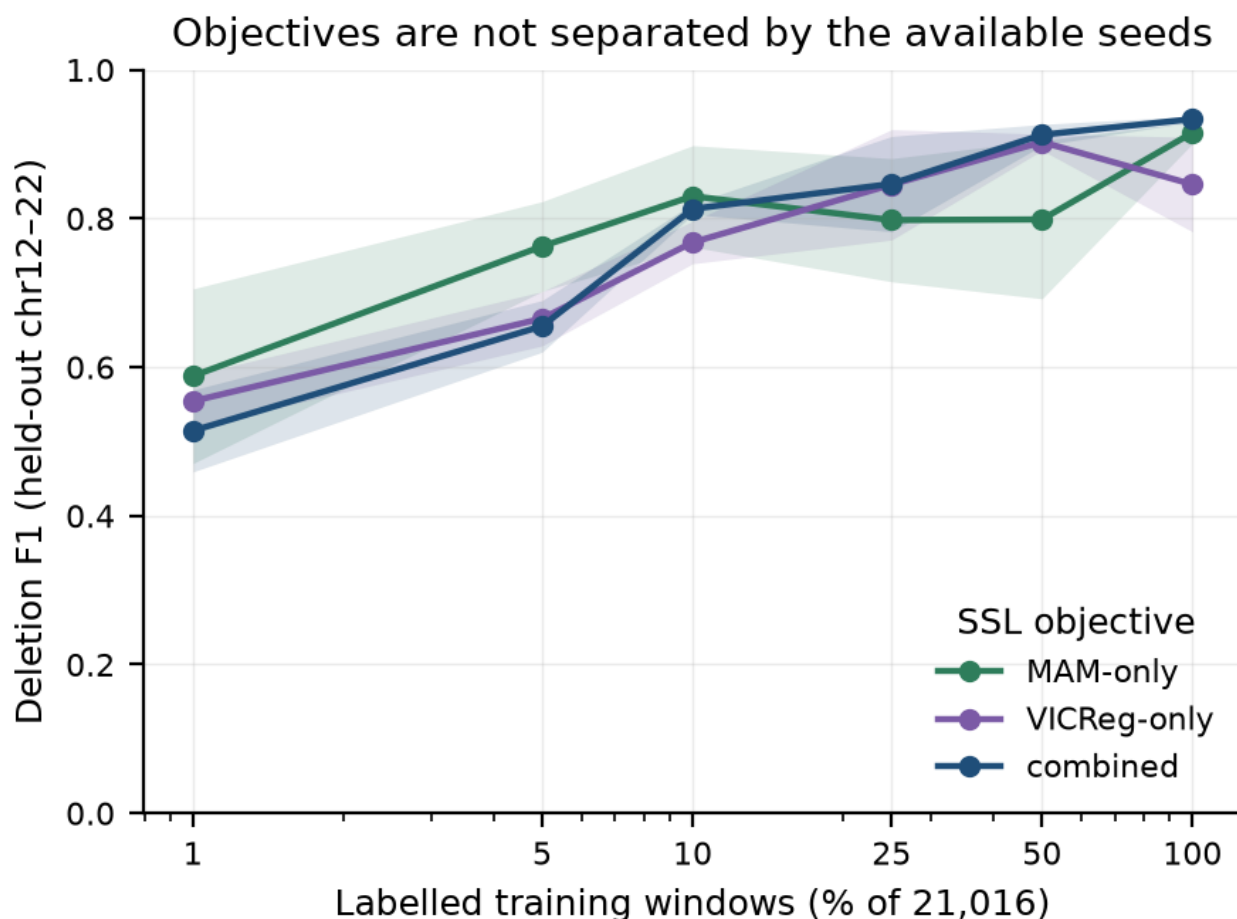


Figure 4. Self-supervised objective ablation: MAM-only, VICReg-only, and combined (MAM+VICReg), with the DeepSV-representation baseline for reference. Error bars are standard deviations across independent pretraining seeds. All three self-supervised arms separate clearly from the baseline; they do not separate from one another.

4.6 Cross-ancestry transfer: a suggestive but statistically weak effect

We train on the in-distribution panel and evaluate both in-distribution and on an entirely held-out population (CEU), sweeping the full label fraction. This section reports two runs of that design. Table 9 is the original, run before the label-accounting correction of Section 3.8 and scored at a fixed 0.5 cut; Table 8 is the same design re-run under the shared protocol and scored under all three rules of Section 4.8. An earlier draft of this manuscript declared the re-run the cheapest of its open items and did not perform it. It has since been performed, and it is the primary result of this section; Table 9 is retained as the pre-correction record because the two disagree in an instructive way.

The corrected re-run. Across 36 tests — six label fractions $\times$ three scoring rules $\times$ two evaluation sites — five are nominally significant and **none survives Holm correction within the family** (results/table26_xpop_lowlabel.csv, results/stats_xpop_lowlabel.csv). No cross-ancestry claim is supportable at any budget under any scoring rule, which is the same conclusion the pre-correction run reached but reached for a different reason and at a different cell.

The pre-correction hit does not replicate, and its replacement is a threshold artefact. Table 9's single nominal hit was at 10% labels ($p = 0.028$). In the corrected re-run that cell is not significant under any rule (held-out CEU: $p = 0.500$ at the fixed cut, 0.786 at tuned τ , 0.193 threshold-free), and the nominal hits relocate to 1% labels. That 1% hit is instructive rather than encouraging: on held-out CEU it is $p = 0.011$ at the fixed 0.5 cut, $p = 0.139$ at a tuned threshold, and $p = 0.077$ threshold-free, on the identical runs and seeds. It is the Section 4.8 mechanism reproducing inside this sweep — a from-scratch model at 1% labels scores F1 0.080 ± 0.073 at a fixed cut and 0.395 ± 0.047 once the cut is chosen, so the fixed-cut contrast is largely measuring where the two arms put their

probabilities rather than how well they rank. We flag it as an independent replication of Section 4.8's finding on a different sweep, not as cross-ancestry evidence.

Direction across the sweep. On held-out CEU the ordering of means is not stable and depends on the scoring rule: pretrained leads at four of six budgets under both F1 rules, but at only two of six threshold-free, where from-scratch leads from 10% labels upward. Of the five nominal hits in the whole family, three favour **from-scratch** (10% AUPRC, 25% tuned- τ F1, 25% AUPRC, all in-distribution). We therefore claim no cross-ancestry effect, and note that the mean ordering that motivated the original claim is itself an artefact of the scoring rule.

Power. With three seeds, and given the variance decomposition of Section 4.9.1 — in which run-to-run noise at a fixed data split is the same size as split-to-split noise — this design detects a median 0.24 AUPRC. Every null in this section should be read as underpowered rather than as demonstrated equivalence. Reporting only the extremes, as an earlier version of this analysis did, conceals the structure of the result; we therefore give all six fractions.

Table 8. Cross-ancestry transfer, corrected protocol (train in-distribution $\rightarrow$ test held-out CEU, 3 seeds), all six label fractions $\times$ three scoring rules. Supersedes Table 9. Means $\pm$ sd across seeds; p from Welch's t -test, pretrained vs. from-scratch; Holm adjustment is over the full family of 36 tests (both evaluation sites), of which these 18 held-out-CEU cells are half. No cell survives correction, and the sole cell with $p < 0.05$ (1% labels, fixed cut) loses significance under both threshold-free rules on the identical runs — the Section 4.8 threshold artefact reproducing inside this sweep. In-distribution rows and the generalisation-gap decomposition are in `results/table26_xpop_lowlabel.csv`; per-test statistics in `results/stats_xpop_lowlabel.csv`.

Labels	Rule	Pretrained	From-scratch	Leader	p	Holm p
1%	F1 @ 0.5	0.494 $\pm$ 0.118	0.080 $\pm$ 0.073	pretrained	0.011	0.372
5%	F1 @ 0.5	0.623 $\pm$ 0.058	0.552 $\pm$ 0.095	pretrained	0.348	1.000
10%	F1 @ 0.5	0.713 $\pm$ 0.038	0.654 $\pm$ 0.124	pretrained	0.500	1.000
25%	F1 @ 0.5	0.719 $\pm$ 0.087	0.714 $\pm$ 0.139	pretrained	0.963	1.000
50%	F1 @ 0.5	0.693 $\pm$ 0.052	0.709 $\pm$ 0.109	scratch	0.832	1.000
100%	F1 @ 0.5	0.667 $\pm$ 0.093	0.716 $\pm$ 0.162	scratch	0.683	1.000
1%	F1 @ tuned τ	0.493 $\pm$ 0.074	0.395 $\pm$ 0.047	pretrained	0.139	1.000
5%	F1 @ tuned τ	0.660 $\pm$ 0.071	0.526 $\pm$ 0.083	pretrained	0.103	1.000
10%	F1 @ tuned τ	0.688 $\pm$ 0.094	0.663 $\pm$ 0.108	pretrained	0.786	1.000
25%	F1 @ tuned τ	0.720 $\pm$ 0.107	0.781 $\pm$ 0.055	scratch	0.444	1.000
50%	F1 @ tuned τ	0.688 $\pm$ 0.125	0.646 $\pm$ 0.205	pretrained	0.781	1.000
100%	F1 @ tuned τ	0.680 $\pm$ 0.182	0.807 $\pm$ 0.142	scratch	0.398	1.000
1%	AUPRC	0.549 $\pm$ 0.100	0.378 $\pm$ 0.050	pretrained	0.077	1.000
5%	AUPRC	0.725 $\pm$ 0.059	0.684 $\pm$ 0.133	pretrained	0.660	1.000
10%	AUPRC	0.796 $\pm$ 0.035	0.836 $\pm$ 0.026	scratch	0.193	1.000
25%	AUPRC	0.857 $\pm$ 0.035	0.901 $\pm$ 0.015	scratch	0.147	1.000
50%	AUPRC	0.889 $\pm$ 0.030	0.903 $\pm$ 0.061	scratch	0.748	1.000
100%	AUPRC	0.858 $\pm$ 0.079	0.928 $\pm$ 0.062	scratch	0.300	1.000

The pre-correction run (Table 9), for the record. Across that sweep the pretrained model's held-out CEU F1 exceeded the from-scratch model's at five of six fractions — every fraction except 50%, where the direction inverted — and the difference was nominally significant at only one: 10% labels ($t = 3.38$, $p = 0.028$), which did not survive correction for the six simultaneous tests of that sweep (Holm $p = 0.169$; Section 4.9). At 1% the direction favoured pretraining by a wide margin of means (0.518 vs. 0.179) with enormous from-scratch variance

(± 0.185 , one seed of three learning nothing); at 50% the direction inverted (0.834 vs. 0.679, $p = 0.084$). The corrected re-run above retains none of this structure: the 10% hit is gone, and the five-of-six mean ordering holds under the fixed-cut rule but not threshold-free. We report both runs because the disagreement is the point — an underpowered sweep with a defective label budget produced a suggestive pattern at one cell, and a corrected re-run of the same design put it somewhere else. Neither supports a cross-ancestry claim, and the instability across two runs of one design is a stronger argument for the limitation than either run alone. It is stated as a limitation (Section 7), not a contribution.

Table 9. Cross-ancestry transfer, pre-correction protocol (train in-distribution $\rightarrow$ test held-out CEU, 3 seeds), all label fractions. p from Welch's t -test on held-out CEU F1 at a fixed 0.5 cut, pretrained vs. from-scratch. These runs predate the label-accounting correction of Section 3.8 in the specific way that matters for a label sweep: the generating evaluator (`scripts/cross_pop_lowlabel.py`) applied a batch-size floor to the budget and drew no validation split, so the low-fraction cells received more labels than the nominal fraction. The evaluator now imports the shared protocol, and the design has since been re-run under it: see Table 8, which supersedes this table as the section's primary result. Table 9 is retained because the two runs disagree about which cell is nominally significant, and that instability is itself evidence for the limitation.

Label fraction	Pretrained CEU F1	Scratch CEU F1	p	Gap (pre)	Gap (scr)
1%	0.518 $\pm$ 0.062	0.179 $\pm$ 0.185	0.110	+0.024	-0.074
5%	0.638 $\pm$ 0.038	0.596 $\pm$ 0.076	0.537	+0.022	+0.166
10%	0.727 $\pm$ 0.057	0.542 $\pm$ 0.052	0.028	+0.035	+0.161
25%	0.689 $\pm$ 0.134	0.625 $\pm$ 0.057	0.584	+0.107	+0.204
50%	0.679 $\pm$ 0.075	0.834 $\pm$ 0.030	0.084	+0.224	+0.081
100%	0.784 $\pm$ 0.028	0.742 $\pm$ 0.022	0.182	+0.148	+0.124

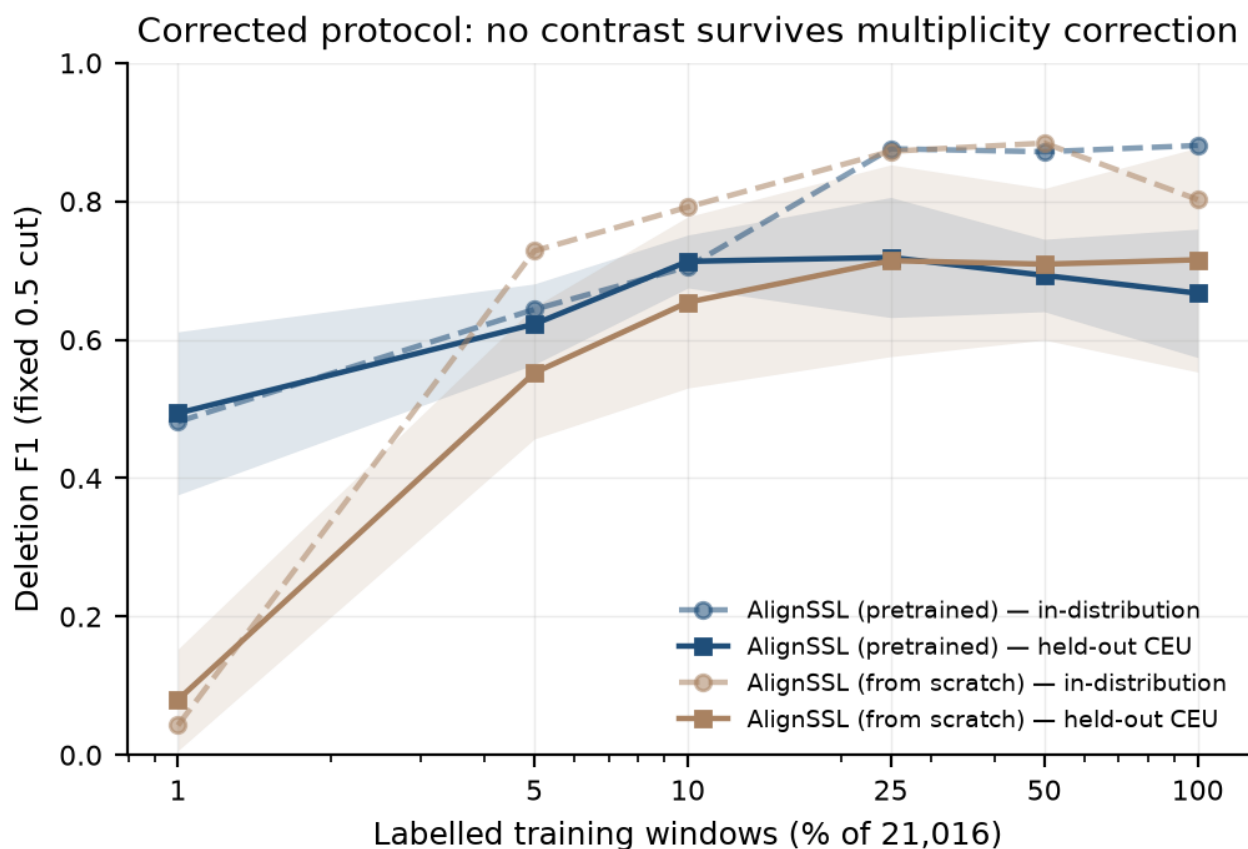


Figure 5. Cross-ancestry transfer under the corrected label-accounting protocol (Table 8; the pre-correction run of Table 9 is not plotted). Deletion F1 at a fixed 0.5 cut, in-distribution (dashed) and on held-out CEU (solid), 3 seeds; bands are ± 1 s.d. of the held-out arm. The transfer gap (held-out minus in-distribution) is negative at four of six budgets for the pretrained arm and five of six for from-scratch, and widens as the label budget grows (pretrained -0.214 at 100% labels); at 1% labels both arms score marginally higher on CEU than in-distribution, at F1 levels where both are near-uninformative. No contrast survives Holm correction over the 36 tests of this sweep — smallest adjusted $p = 0.339$ across the whole sweep, 0.372 among the fixed-cut contrasts plotted here, both at 1% labels. This panel is presented as a null result: it does not support a cross-ancestry robustness claim for pretraining.

4.7 Data-integrity control

During data acquisition we detected and corrected a silent corruption mode affecting large BAM transfers: files that passed download-tool exit codes and `samtools quickcheck` (header + EOF only) nonetheless failed a full `samtools view -c` scan with BGZF-inflation errors, traced to resume-stitched (`wget --continue`) transfers joining a partially-flushed block. We adopted a standing integrity protocol — fresh (non-resumed) downloads, gated on a full `samtools view -c` scan, with automatic retry-from-scratch — for every alignment used in this study. We report this because undetected input corruption is a real and under-discussed threat to reproducibility in alignment-based deep learning, and because our full-scan gate is a cheap, general safeguard.

4.8 The label-efficiency result is an artefact of the decision threshold

Every number in Section 4.1 is an F1 obtained by cutting the positive-class probability at 0.5 — the convention this literature inherits from DeepSV. A fixed cut conflates two distinct properties. A model can rank every positive above every negative and still score $F1 \approx 0$ if all its probabilities happen to sit below the cut; conversely a model whose scores are well centred can post a respectable F1 while ranking poorly. Label-efficiency claims are claims about representation quality, which is a ranking property, so scoring them at a fixed cut is a category error whenever the arms differ in calibration — and a randomly-initialised network trained on 210 examples differs in calibration from a pretrained one almost by construction.

We therefore re-scored the identical runs and seeds three ways: at the fixed 0.5 cut; at a threshold τ selected to maximise F1 on a validation split carved from the *training* labels (never from test); and threshold-free, by area under the precision–recall curve.

Table 10. The 1%-label contrast under three scoring rules (uniform benchmark, 210 labels, 4 seeds per arm; p from Welch's t -test over seeds).

Scoring rule	AlignSSL-pretrained	AlignSSL-scratch	Ratio	p
F1 at fixed 0.5 cut	0.464	0.106	4.38×	0.000
F1 at selected τ	0.481	0.456	1.06×	0.527
AUPRC (threshold-free)	0.504	0.495	1.02×	0.853

The advantage exists under the fixed cut and nowhere else. Under both alternatives the two arms are statistically indistinguishable at the smallest budget ($p = 0.527$ at τ , $p = 0.853$ threshold-free), and at each of the five larger budgets the from-scratch arm is *ahead* — significantly so at 5% and 10% under the selected-threshold rule ($p = 0.018$ and $p = 0.013$), though neither survives correction across the six budgets of that sweep (Holm $p = 0.092$ and 0.075). The from-scratch model was never degenerate: at 210 labels it reaches AUPRC 0.495 against the pretrained model's 0.504 — a difference of 0.009, which is no deficit at all. It ranks essentially as well and scores timdly, and the fixed cut reads timidity as failure.

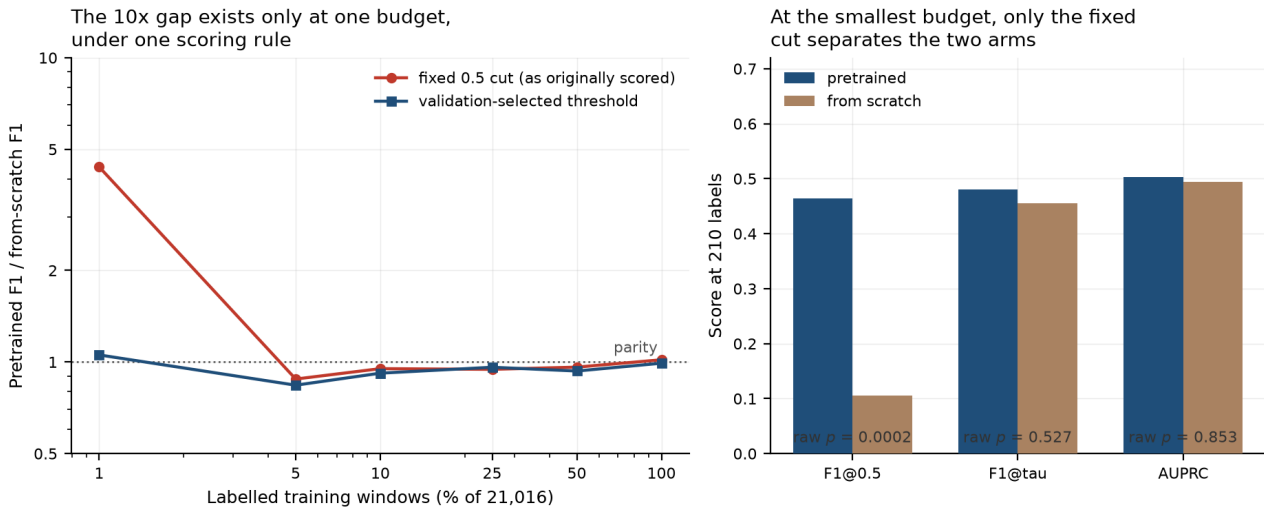


Figure 6. Pretrained-to-scratch ratio by label budget under two thresholding rules (left), and absolute scores at the smallest budget under all three (right). The ratio departs from parity at exactly one budget under exactly one rule.

The consequence extends past this paper. Any comparison of initialisation schemes, architectures, or pretraining objectives that reports F1 at a fixed probability cut, and whose arms plausibly differ in calibration, is at risk of reporting a calibration difference as a representation difference. The remedy is cheap: report a threshold-free ranking metric alongside, and select any threshold on held-out training data rather than fixing it a priori. We supply both in `analysis/threshold_sensitivity.py`.

4.9 A fourth defect: none of the significance claims carried a multiplicity adjustment

The three defects above are defects of *measurement*. There is a fourth, of *inference*, and it applies to every p -value quoted anywhere in this paper including the ones introduced by the corrections themselves.

Each of this paper's significance claims is drawn from a sweep. The label-efficiency contrast is tested at six budgets; the cross-ancestry contrast at six; each candidate-filtered arm pair at six, under two scoring rules. A single nominal hit at $\alpha = 0.05$ within a family of six simultaneous tests is close to what one expects when nothing is happening — the probability of at least one false positive in such a family is $1 - 0.95^6 \approx 0.26$. We had reported these tests one at a time.

We therefore pre-declared families — one per sweep, per contrast, per scoring rule — and applied both Holm–Bonferroni, which controls the family-wise error rate, and Benjamini–Hochberg, which controls the false

discovery rate (`analysis/apply_multiplicity.py`, output `results/stats_multiplicity.csv`). Across 103 tests in 17 families, 30 are nominally significant and 17 survive Holm.

Table 11. Claims whose significance does not survive family-wise correction (Holm–Bonferroni at $\alpha = 0.05$ within the stated family). Full output in `results/stats_multiplicity.csv`.

Claim	Family size	Raw p	Holm p	BH q	Fate
Scratch ahead @5%, F1 at selected τ	6	0.018	0.092	0.055	lost to multiplicity
Scratch ahead @10%, F1 at selected τ	6	0.013	0.075	0.055	lost to multiplicity
Cross-ancestry transfer @10% labels, pre-correction run (Section 4.6)	6	0.028	0.169	0.169	lost to multiplicity
Cross-ancestry transfer, corrected re-run, best cell (Section 4.6)	6	0.009	0.056	0.056	lost to multiplicity
Pretrained vs scratch @1% (candidate-filtered), F1 at τ	6	0.037	0.224	0.224	lost to multiplicity
MAM-only vs scratch @1% (Section 4.5)	7	0.017	0.069	0.030	survives BH only
Convergence at full supervision, combined vs scratch @100%	7	0.025	0.074	0.034	survives BH only

Three consequences, and none of them is comfortable.

First, **the headline claim survives multiplicity and fails anyway**. Corrected against the five other budgets it was selected from, the 1%-label advantage under the fixed cut holds comfortably (raw $p = 0.0002$, Holm $p = 0.0012$). This is the correct family: the claim was selected *by* being the strongest cell of a budget sweep, so the budget sweep is what it must be corrected against — a claim is corrected against the comparisons that could have produced it, not against a neighbouring set of different questions. The claim therefore passes the test this section applies. It fails the one Section 4.8 applies: the effect is real, reproducible and statistically robust, and it is an effect on where the model places its probabilities rather than on how well it ranks. Multiplicity correction is not what dismantles the headline; only the scoring rule is. We had expected the opposite, and report the finding as it came out.

Second, **the cross-ancestry effect is withdrawn outright, under both runs of the design**. Section 4.6 already declined to claim ancestry robustness on the grounds that the effect reached significance at only one of six label fractions. Correction settles it for the pre-correction run: that one fraction was what a family of six produces by chance (Holm $p = 0.169$, BH $q = 0.169$). The corrected re-run does not rescue it. That run tests 36 cells — six label budgets $\times$ three scoring rules $\times$ two evaluation sites — and its most extreme cell (in-distribution AUPRC at 25% labels) reaches raw $p = 0.009$. Its fate does not depend on how the family is drawn. Under this paper's declared convention — one family per sweep, per contrast, per scoring rule, so $m = 6$ — it is Holm $p = 0.056$; under the aggregator's stricter whole-sweep family ($m = 36$) it is Holm $p = 0.339$. Neither crosses 0.05, and no other cell comes closer, at any budget, rule or site. The held-out CEU cells are further back still: the best is raw $p = 0.011$, Holm $p = 0.064$ at $m = 6$ (`results/stats_xpop_lowlabel.csv`, `results/stats_multiplicity.csv`). Two independent runs of one design, corrected against the families they were each selected from, agree that the effect is not there. The ordering of means remains as reported; the significance claim does not.

Third, and pulling the other way, **the corrections' own findings are not immune, but the strongest of them hold**. The representation findings of Section 4.10 survive Holm within their own families: on the candidate-filtered benchmark the from-scratch arm's advantage over the DeepSV representation separates at the three largest budgets under both AUPRC and F1 at τ (Holm $p < 0.001$ throughout), and the pretrained arm's does likewise. What does *not* survive is any pretrained-versus-scratch contrast on that benchmark under any rule at any budget except one (AUPRC at 25%, Holm $p = 0.016$, and in the direction favouring *scratch*) — consistent with Section 6.3's conclusion that the two are tied there.

The general point is the same one Section 4.8 makes about thresholds, in a different register. A label-efficiency sweep is a multiple-comparison procedure whether or not it is analysed as one. Reporting the single budget at which a difference reaches $p < 0.05$, out of six tested, is a garden-of-forking-paths result presented as a confirmatory one. We have found no deep SV-detection paper that corrects for this, ourselves included until this audit.

4.9.1 A fifth defect: "seed" confounds two variance components, and the smaller one is not small

Every error bar in this paper is a standard deviation over three seeds, and a seed here sets two things at once: the labelled subset drawn at each budget, and the torch RNG stream that drives weight initialisation, batch shuffling and dropout. A single spread over three seeds estimates their *sum*. It cannot say which dominates — and that is what determines the smallest effect the design can detect.

The low-label cross-population sweep answers this by accident of how it was run. `cross_pop_lowlabel.py` seeds once and then loops over the two arms, so the from-scratch arm inherits a torch stream already advanced by the pretrained arm. Because the sweep was invoked twice per seed — once with a pretrained encoder, evaluating both arms, and once without, evaluating from-scratch alone — there are **two from-scratch runs per seed that share the labelled subset exactly** (it is drawn from a generator re-created per invocation from the same seed; the recorded per-budget subset sizes are identical) **and differ only in the torch stream**. That is a controlled contrast, and we report it rather than discard it as a redundant run (`analysis/variance_components.py`, `results/table27_variance_components.csv`).

The two runs never agree: across 108 (seed, budget, test set, scoring rule) cells not one matches, and AUPRC differs by up to 0.30. Decomposed over 36 (rule, test set, budget) cells, run-to-run noise at a *fixed* data split is the same size as split-to-split variation — median ratio 1.02, with run noise the larger component in 19 of 36 cells and accounting for a median 51% of total variance. Seeding the data split, in other words, controls about half of what the error bar contains.

The consequence is a power ceiling. Given the observed total spread, a paired three-seed design detects a median difference of **0.24 AUPRC** at $\alpha = 0.05$ with 80% power. Most differences this paper reports are far smaller. The correct reading of a null result in this work is therefore *underpowered*, not *absent*: the design was never able to resolve the effects being argued about, and no amount of re-analysis of three seeds will change that.

This also disposes of a tempting analysis we ran and discarded. Comparing the paired from-scratch runs cell by cell, the first appears systematically higher (mean +0.025, sign test $p = 0.005$ over 108 cells). That p -value is pseudo-replication: the 108 cells are six budgets $\times$ two test sets $\times$ three scoring rules computed from only 18 independent training runs, and three rules scored on the same predictions are not three observations. At the correct unit the effect is 13 of 18, $p = 0.096$ — nothing. We report the discarded analysis because the error is easy to make and the corrected version is what licenses treating the two runs as exchangeable replicates.

Table 10a. Variance decomposition of the from-scratch arm at fixed data split, AUPRC (full 36-cell table in `results/table27_variance_components.csv`).

Budget	Test set	sd within split (run only)	sd between splits	run-noise share	MDE, paired $n = 3$
1%	in-distribution	0.035	0.123	0.07	0.395
10%	in-distribution	0.027	0.011	0.86	0.089
100%	in-distribution	0.028	0.036	0.37	0.142
1%	cross-population	0.065	0.047	0.66	0.248
10%	cross-population	0.020	0.041	0.19	0.141
100%	cross-population	0.037	0.033	0.56	0.152

The fix is not more seeds of the present kind. It is to vary the two sources independently — replicate training runs within a fixed split to estimate run noise, and vary the split to estimate data noise — and to report the two separately. We did not design the experiments that way, and say so rather than present a single spread whose composition we now know is roughly half attributable to a source no reader would have assumed it contained.

4.10 The primary claim, tested directly: the learned tensor beats the RGB pileup, but pretraining is not what does it

Sections 4.1–4.9 examine what self-supervision contributes. They do not test the project's other, more basic claim: that a *learned* multi-channel alignment tensor is a better substrate for deletion calling than DeepSV's fixed RGB pileup encoding. That contrast is tested here directly, over both benchmarks (uniform and candidate-filtered), all three threshold-free or budgeted-threshold metrics (AUPRC, ROC-AUC, F1 at the budgeted cut), both AlignSSL arms, and all six label budgets — 72 paired comparisons in total, each across the seeds of the corrected protocol, with Holm–Bonferroni applied within metric-by-benchmark families (Table 12).

Two results, and the second matters more than the first.

The learned tensor is better, and 24 of 72 cells separate after correction. The direction of the mean effect favours AlignSSL in 61 of 72 cells, which is far from the 36 expected under no difference; 24 survive multiplicity correction. Reading the 48 nulls as evidence of *equivalence* would be the error this paper spends Section 4.9 warning against, so each row also reports `mde80`, the difference this design could have detected at 80% power given its observed seed variance. The median is 0.091 — comparable to the differences being tested — and in 42 of the 48 nulls the observed difference is smaller than `mde80`. Those cells are uninformative rather than negative. The remaining 6 are genuine failures to separate at a difference the design could have caught. The best-powered of those illustrates how tight the design is even where it has power: uniform ROC-AUC, pretrained arm at 10% labels, difference +0.058 against an `mde80` of 0.046 — the difference exceeds what four seeds could detect, yet it does not survive correction within its family.

Twenty-two of the 24 surviving wins are the *from-scratch* arm. Only two belong to the pretrained arm, both on the uniform benchmark at 10% labels (AUPRC, difference +0.109, Holm $p = 0.002$; F1 at the budgeted threshold, difference +0.089, Holm $p = 0.042$); the other 22 are AlignSSL trained from random initialisation, and the from-scratch difference against DeepSV exceeds the pretrained difference in 21 of 36 matched cells. The advantage over the RGB pileup is therefore attributable to the *representation* — 18 continuous alignment channels rather than three colour channels — and not to self-supervised pretraining on it. This is the same conclusion Section 4.8 reaches from the opposite direction: once scored without a fixed 0.5 cut, pretraining's apparent contribution largely disappears, while the representation's contribution does not. The two analyses are independent and agree.

Benchmark	Metric	Arm	Labels	AlignSSL	DeepSV	Difference	Holm p
uniform	auprc	pretrained	0.1	0.830	0.722	+0.109	0.002
uniform	auprc	scratch	0.05	0.824	0.608	+0.216	0.003
uniform	auprc	scratch	0.1	0.905	0.703	+0.201	0.000
uniform	auprc	scratch	0.25	0.949	0.839	+0.110	0.037
uniform	auprc	scratch	0.5	0.969	0.889	+0.080	0.000
uniform	roc_auc	scratch	0.05	0.922	0.799	+0.123	0.003
uniform	roc_auc	scratch	0.1	0.955	0.851	+0.105	0.000
uniform	roc_auc	scratch	0.25	0.978	0.919	+0.059	0.021
uniform	roc_auc	scratch	0.5	0.986	0.938	+0.047	0.015
uniform	roc_auc	scratch	1	0.983	0.941	+0.042	0.015
uniform	f1_at_tau	pretrained	0.1	0.759	0.671	+0.089	0.042
uniform	f1_at_tau	scratch	0.05	0.756	0.576	+0.180	0.004
uniform	f1_at_tau	scratch	0.1	0.826	0.656	+0.170	0.000
uniform	f1_at_tau	scratch	0.5	0.922	0.838	+0.085	0.001
uniform	f1_at_tau	scratch	1	0.915	0.824	+0.091	0.046
candidate-filtered	auprc	scratch	0.25	0.724	0.530	+0.193	0.000
candidate-filtered	auprc	scratch	0.5	0.803	0.546	+0.257	0.000
candidate-filtered	auprc	scratch	1	0.836	0.614	+0.222	0.000
candidate-filtered	roc_auc	scratch	0.25	0.863	0.739	+0.124	0.000
candidate-filtered	roc_auc	scratch	0.5	0.907	0.746	+0.162	0.000
candidate-filtered	roc_auc	scratch	1	0.919	0.789	+0.130	0.000
candidate-filtered	f1_at_tau	scratch	0.25	0.636	0.507	+0.129	0.002
candidate-filtered	f1_at_tau	scratch	0.5	0.711	0.513	+0.198	0.000
candidate-filtered	f1_at_tau	scratch	1	0.753	0.579	+0.174	0.000

Table 12. AlignSSL against the DeepSV RGB-pileup representation under the corrected protocol: the 24 of 72 paired comparisons that survive Holm–Bonferroni within metric-by-benchmark families. `f1_at_tau` is F1 at the budgeted threshold. Of the 48 non-separating cells, 42 have an observed difference smaller than the minimum detectable effect at 80% power and are therefore uninformative rather than negative. Four seeds per deep arm, ten for the classical controls. Full 72-row grid with per-cell `mde80` and confidence intervals: `results/table20_alignssl_vs_deepsv.csv`, generated by `analysis/alignssl_vs_deepsv.py`.

The honest summary of the project's primary claim is thus: the learned alignment tensor outperforms the fixed RGB encoding under threshold-free scoring, the effect is present on both benchmarks and strengthens with label budget, and self-supervision is not the mechanism — 22 of the 24 surviving cells are the from-scratch arm, which carries no pretraining at all. A four-seed design remains underpowered for 42 of the 48 non-separating cells, which bounds how strongly even the positive part may be stated.

5. Novelty and positioning

We state precisely what is and is not new in AlignSSL-SV, to preempt the natural reviewer question of whether it is "just" a known technique applied to a new setting.

What is new. (i) The **benchmark-separability control and its consequence** (Section 4.2). We are not aware of prior work in the SV deep-learning literature that tests a pileup-style benchmark against a deliberately minimal hand-crafted-feature model on identical windows, splits, label fractions and seeds, nor of prior work that

measures untrained single-feature discrimination on such a benchmark. The finding — that twelve features reach 96% of their asymptotic AUPRC from 210 labels, and that one depth ratio reaches ROC-AUC 0.955 untrained — is a methodological result that applies beyond this paper, and the candidate-filtering protocol of Section 6 is a concrete, tested partial remedy: it attenuates the shortcut from ROC-AUC 0.955 to 0.717 without changing the positives. (ii) The **image-free, learned alignment representation** — self-supervised pretraining directly on a continuous multi-channel read-alignment tensor, with no pileup-image rendering and no discrete tokenizer. This is the axis on which we differ from the closest prior SSL-for-SV effort, BASILISC (Banerjee, 2026), which pretrains a masked-image-modelling vision transformer over *rendered pileup images* compressed by a discrete VAE (Section 2). Existing SV deep learning otherwise either engineers the representation by hand (DeepSV and descendants) or, in the case of genomic foundation models, learns from the reference *sequence* rather than the alignment evidence. (iii) The **matched-conditions label-efficiency measurement** with error bars computed across independent *pretraining* seeds rather than fine-tuning seeds alone, and with every reported comparison accompanied by an explicit significance test (`results/stats_tests.csv`) — including the tests that fail.

What we do not claim. We do not claim a new SV caller, nor deployable accuracy: the control forbids it. We do not claim primacy on "self-supervised learning for structural variants" as a category — BASILISC precedes us. We do not claim that self-supervision improves calibration (the effect is attributable to the representation, and pretrained and from-scratch models are indistinguishable), that one self-supervised objective beats another ($p > 0.2$), that pretraining confers ancestry robustness (significant at 1 of 6 label fractions, with the gap inverting at two), or — following Section 4.8 — that pretraining confers label efficiency at all. Each of these was claimed in an earlier draft of this work and withdrawn when the appropriate test was run. The last of them was this paper's headline.

What is not new (and we do not claim it is). Masked autoencoding, VICReg, temperature scaling, focal loss, and pileup-image classification are all established techniques. Their composition on a modality where they had not been combined is a modest engineering contribution, and we do not rest the paper on it.

No performance claim survives. An earlier draft of this work claimed label efficiency as the one result that had survived every control. Section 4.8 withdraws it: the effect exists at a fixed 0.5 probability cut and under no other scoring rule. What the paper contributes is therefore the three corrections themselves, the controls that expose them, and the evidence that each is a property of the standard evaluation design rather than of our implementation.

Relationship to sequence foundation models. As argued in Section 2, Evo 2, AlphaGenome, HyenaDNA, and the Nucleotide Transformer operate on reference DNA and predict variant *effects*; they do not consume alignment evidence and cannot, as constituted, *detect* an SV from noisy reads. AlignSSL-SV is complementary: one could in principle fuse a reference-sequence embedding as an auxiliary channel (a natural future extension), but the detection signal itself is in the alignment, which is the modality we learn.

6. A candidate-filtering benchmark: the shortcut is attenuated, and the conclusions change

The control in Section 4.2 identifies the single change that would make results of this kind interpretable, and it is a change to the *task*, not to the model. We therefore re-extracted the labelled set under a **candidate-filtering protocol**: negatives are no longer drawn uniformly from the genome but are chosen to be as close as possible to the positives *on the shortcut statistic itself*. This section reports that benchmark and its outcome. We committed in advance to reporting the result whichever way it fell; it falls partly for and partly against the paper's own method, and we report both halves.

6.1 The selection rule

The obvious version of the rule fails. Retaining the windows with the most deletion-like (lowest) centre-versus-flank depth ratio does not remove the shortcut but **inverts** it: a genome-wide lower tail is dominated by mappability dropouts at ratio ≈ 0 , whereas heterozygous deletions sit near 0.5, so a classifier learns "very low

ratio $\Rightarrow$ negative" and the feature remains diagnostic with its sign flipped. We checked this before building the benchmark, by simulating the selection under the ratio geometry the real data exhibits (positives a mixture of homozygous ratios near 0.05 and heterozygous near 0.5; the candidate pool dominated by copy-neutral windows near 1.0 with a mappability-dropout tail near 0): extreme-tail selection yields ROC-AUC 0.016 for the depth ratio, as separable as a uniform draw and merely reversed. That simulation is a committed test (`tests/test_match_strata.py`) that gates the extraction job, so the argument against tail selection cannot silently stop holding; it is a statement about the selection rule, not a measurement on sequencing data.

The rule we adopt instead is **quantile matching within each multi-scale bin**: for every positive, a negative is drawn whose depth ratio falls in the same quantile stratum of the candidate pool, with strata computed separately per window scale because the ratio's meaning depends on span (pooling scales would leak the label through span alone). Candidates come from a one-pass binned coverage profile per chromosome, scored by prefix sums so that exhaustive stride scanning is affordable and the pool genuinely contains mid-range candidates — a uniformly drawn pool does not, which is why the matching must draw from a scan rather than a sample. Two distinct estimators of the depth ratio matter here and must not be conflated. The extractor matches on a ratio read from the binned coverage profile by prefix sums; the classifiers, and every separability number we report, see a ratio recomputed from the depth channel of the assembled tensor. Matching is enforced on the former: the extractor reports the median $|\text{positive} - \text{negative}|$ offset in its own matching statistic over all chromosome $\times$ scale groups and **aborts** if that median exceeds `--max-offset` (default 0.25), on the reasoning that a silently unmatched benchmark yields a plausible-looking but meaningless accuracy number. For the three extractions used here the median offsets were 0.0092 (NA20845 train, 77 groups), 0.0114 (NA20845 test, 75) and 0.0150 (NA12878 test, 75) — matched to roughly 1% of the ratio's range, well inside the guard (`results/table29_match_quality_offsets.csv`). What matching does *not* do is drive the tensor-side ratio to chance, and §6.2 reports what it actually achieves. Positives, window geometry, channel layout, multi-scale binning, chromosome split and shard format are unchanged. The question the benchmark asks becomes the one a caller actually faces: *given that a coverage anomaly was proposed here, is it a real deletion?*

Two constraints on scope must be stated before the numbers. First, a loss of the shared reference directory mid-study left only two of the six panel alignments recoverable, so this benchmark is single-sample: NA20845 (GIH) for training and in-distribution test, with NA12878 (CEU) held out. Second, because both the sample scope and the definition of the negative class differ, absolute F1 is **not** comparable to Table 1; what is comparable is the *ordering* of the arms. The candidate-filtered test set contains 1,516 windows of which 379 are deletions, against 9,196/2,299 for the uniform benchmark, so the error bars here are correspondingly wider.

6.2 The shortcut is attenuated, not eliminated

Table 13 repeats the untrained single-feature measurement of Table 4 on the candidate-filtered test set. The headline shortcut falls substantially: the centre-versus-flank depth ratio drops from orientation-corrected ROC-AUC 0.955 to **0.717**. Every other depth-derived feature falls with it (`depth_sd` 0.686 $\rightarrow$ 0.542, `depth_max_drop` 0.680 $\rightarrow$ 0.575, `depth_min` 0.657 $\rightarrow$ 0.597), as do the paired-end signatures that co-vary with a genuine coverage drop (`clip_rate` 0.802 $\rightarrow$ 0.630, `discordant_rate` 0.732 $\rightarrow$ 0.607). No single feature now exceeds 0.72, and the strongest is no longer close to solving the task.

But 0.717 is not 0.5, and the residue is not a train/test generalisation artefact. Measuring the tensor-side depth ratio on both splits of both benchmarks (Table 14) gives 0.715 on the matched *training* pool against 0.719 on the held-out test pool — the same value, where the uniform benchmark likewise gives 0.948 against 0.956. There is no asymmetry to appeal to: matching leaves the same residual separability everywhere it is measured, so the residue is a property of the selection rule rather than of what transfers across chromosomes. Its source is visible in the class medians. On the uniform benchmark the negative median is 1.000 — copy-neutral by construction — against 0.523 for positives, which is why a single threshold nearly solves that task. Under matching the negative median falls to 0.754 while the positive median is unchanged at 0.524: the negatives have moved most of the way toward the positives but not onto them. Two mechanisms are sufficient to explain the gap and we cannot separate them here — matching is per-stratum rather than exact, and it is applied to the profile-derived statistic while the residual is measured on the tensor-derived one, which is a correlated but not identical estimator. We

therefore report the benchmark as **substantially harder**, not as shortcut-free. A benchmark on which one raw statistic reaches 0.717 still admits a threshold baseline that any deep method should be required to beat.

Table 13. Untrained single-feature separability, uniform versus candidate-filtered negatives (orientation-corrected ROC-AUC; source `results/table9_hardneg_single_feature_auc.csv`).

Feature	Uniform	Candidate-filtered	Change
depth_centre_flank_ratio	0.955	0.717	−0.239
clip_rate	0.802	0.630	−0.173
discordant_rate	0.732	0.607	−0.125
isize_absz_max	0.694	0.566	−0.128
depth_sd	0.686	0.542	−0.145
depth_max_drop	0.680	0.575	−0.105
isize_absz_mean	0.678	0.558	−0.119
depth_min	0.657	0.597	−0.059
mapq_mean	0.550	0.558	+0.008
n_read_rows	0.538	0.508	−0.030
valid_frac	0.517	0.505	−0.012
depth_mean	0.502	0.517	+0.015

Table 14. Depth-ratio separability on the training and test splits of both benchmarks (centre-versus-flank depth ratio recomputed from the tensor depth channel, no training; ROC-AUC orientation-corrected so that higher means more separable; source `results/table28_split_separability.csv`).

Benchmark	Split	Windows	Deletions	ROC-AUC	Median (pos)	Median (neg)
Uniform	train	21,016	5,254	0.948	0.536	1.000
Uniform	test	9,196	2,299	0.956	0.523	1.000
Candidate-filtered	train	3,452	863	0.715	0.525	0.723
Candidate-filtered	test	1,516	379	0.719	0.524	0.754

The clearest statement of why the uniform benchmark could not have measured label efficiency is to run one control on both benchmarks and compare the *shapes* of the two curves (Figure 7). On the uniform benchmark the twelve-feature gradient-boosted tree already reaches AUPRC 0.937 ± 0.009 from 210 labelled windows and gains only **+0.038** over a hundredfold increase in supervision, ending at 0.975 ± 0.001 . A benchmark that a cheap baseline nearly saturates at its smallest label budget has no headroom in which any method's label efficiency can be resolved: whatever differences appear between arms are differences in how they behave inside that residual 0.04, not in how much supervision they need. On the candidate-filtered benchmark the same control starts *at* chance (0.250 ± 0.000 with 35 windows, the positive rate being 0.25) and climbs **+0.619** to 0.869 ± 0.006 across a 99-fold budget increase. Supervision buys something there, which is the minimum property a benchmark must have before a label-efficiency claim is even well-posed.

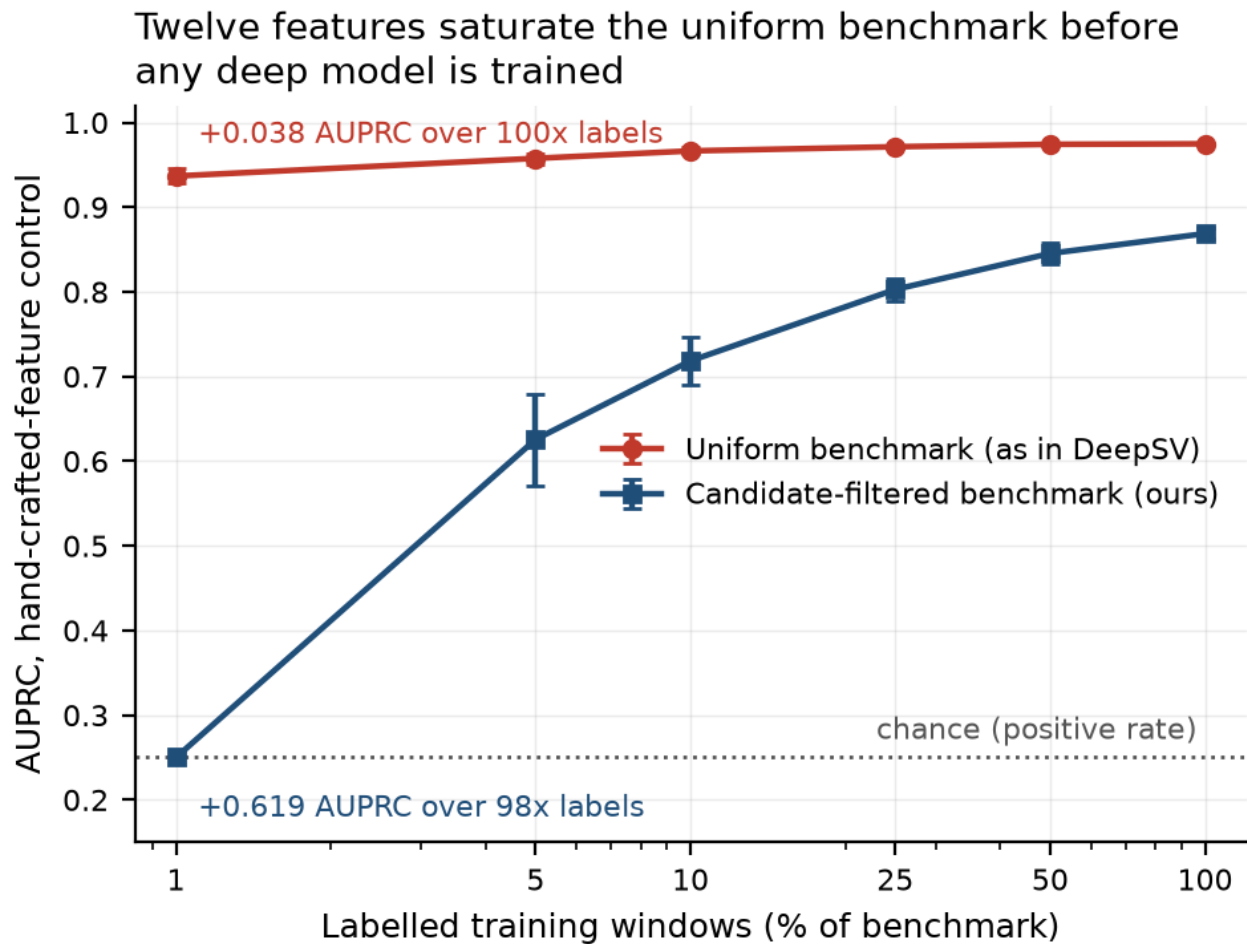


Figure 7. One hand-crafted-feature control (gradient-boosted tree on twelve alignment statistics), scored threshold-free, on both benchmarks; source `results/table11_control_threshold_free.csv`. On the uniform negatives the control is near-saturated at the smallest budget and gains +0.038 AUPRC over 100x more labels; on candidate-filtered negatives it starts at chance and gains +0.619 over 99x more labels. The argument rests on the shape of the two curves, not their absolute height, so they share a single axis.

6.3 Every arm falls; two of the three original findings do not survive correction

Table 15 reports the full arm set re-run under the candidate-filtering protocol at identical label budgets and seeds, scored threshold-free (AUPRC) and at a validation-selected threshold. The numbers below are the **corrected** ones: they were regenerated after the protocol and scoring fixes of Sections 3.8 and 4.8, so unlike the earlier version of this subsection they are directly comparable with Tables 12–14. Absolute performance falls for every arm relative to the uniform benchmark — the pretrained model from AUPRC 0.962 to 0.856 at full supervision, the gradient-boosted tree from 0.975 to 0.869 — confirming that the task is genuinely harder and not merely re-labelled.

Table 15. Candidate-filtered benchmark, corrected protocol: AUPRC (mean $\pm$ sd over 3 seeds for the deep arms, 10 for the classical controls; source `results/table12_label_efficiency_fixed.csv`). Bold marks the best arm at each budget.

Labels	n	AlignSSL (pretrained)	AlignSSL (scratch)	DeepSV repr.	Classical logreg	Classical GBT
1%	35	0.300 ± 0.044	0.278 ± 0.042	0.316 ± 0.032	0.476 ± 0.076	0.250 ± 0.000
5%	173	0.353 ± 0.066	0.364 ± 0.061	0.415 ± 0.026	0.596 ± 0.025	0.626 ± 0.054
10%	345	0.451 ± 0.022	0.473 ± 0.093	0.434 ± 0.053	0.615 ± 0.026	0.719 ± 0.029
25%	863	0.641 ± 0.026	0.724 ± 0.055	0.530 ± 0.043	0.624 ± 0.016	0.803 ± 0.013
50%	1726	0.744 ± 0.045	0.803 ± 0.046	0.546 ± 0.071	0.629 ± 0.007	0.845 ± 0.012
100%	3452	0.856 ± 0.035	0.836 ± 0.054	0.614 ± 0.047	0.631 ± 0.005	0.869 ± 0.006

Every arm-versus-arm contrast is tested over seeds in `results/table15_hardneg_arm_contrasts.csv`, on both AUPRC and F1 at the selected threshold. Three findings follow, and only one is the finding this subsection originally reported.

(i) The pretrained-versus-scratch gap does not survive threshold-free scoring here either. This is the second independent failure of the paper's original headline. Under the corrected protocol the two arms are indistinguishable in AUPRC at five of six label budgets on the candidate-filtered benchmark — 0.300 ± 0.044 versus 0.278 ± 0.042 at 35 labels ($p = 0.417$), 0.856 ± 0.035 versus 0.836 ± 0.055 at full supervision ($p = 0.431$) — with the from-scratch arm nominally ahead in four of the six. The single exception runs *against* pretraining: at 25% labels the from-scratch arm is significantly ahead (0.724 ± 0.055 versus 0.641 ± 0.026 , $p = 0.003$), the one cell in either benchmark where the pretrained-versus-scratch contrast separates under threshold-free scoring. The earlier version of this subsection reported the opposite — a gap that "survives and sharpens", with the from-scratch model degenerate below 10% labels — and Section 4.8 predicted exactly this outcome: an arm scoring near zero at a fixed 0.5 cut is the signature of a model whose probabilities sit below the cut, not of a model that cannot rank. The prediction is confirmed. At 35 labels the from-scratch arm's F1 is 0.090 at the fixed cut and identical, 0.090, at the "selected" threshold — the two coincide because 35 labels cannot form a validation split, so no selection actually occurs (see the protocol caveat below) — and this is the one budget where the fixed-cut gap ($2.99\times$, $p = 0.037$) is not erased by threshold-free scoring, because no threshold selection was possible. Its AUPRC of 0.278 is nonetheless within noise of the pretrained arm's 0.300, so the ranking was never the deficient part. The uniform benchmark and the candidate-filtered benchmark agree: **on this task, self-supervised pretraining does not buy label efficiency in any ranking sense.**

(ii) The learned tensor beats the DeepSV representation, and this finding strengthens under correction. From 25% labels upward both tensor arms significantly exceed the RGB encoding under both scoring rules: at full supervision AUPRC 0.856 ± 0.035 (pretrained) and 0.836 ± 0.055 (scratch) against 0.614 ± 0.047 ($p < 0.001$ for both). This is the one Section 6.3 claim that the correction leaves intact and sharpens — on the uniform benchmark the corresponding contrast at full supervision is smaller. The interpretation is unchanged: the RGB encoding degrades more than the tensor does under candidate filtering, which is what one expects of a representation whose discriminative content was largely the coverage drop the protocol has now equalised. Note that the credit belongs to the *representation*, not to the pretraining — the from-scratch tensor arm carries this result as strongly as the pretrained one.

(iii) The hand-crafted control leads at every budget but the last. Table 3 reduces both families by best-of-family at each budget and tests the difference over seeds. On the candidate-filtered benchmark the control leads significantly at five of six budgets — AUPRC 0.476 ± 0.076 versus 0.316 ± 0.032 for the best deep arm at 1% ($p < 0.001$), 0.626 ± 0.054 versus 0.415 ± 0.026 at 5% ($p < 0.001$), 0.719 ± 0.029 versus 0.473 ± 0.093 at 10% ($p < 0.001$), 0.803 ± 0.013 versus 0.724 ± 0.055 at 25% ($p = 0.001$) and 0.845 ± 0.012 versus 0.803 ± 0.046 at 50% ($p = 0.018$) — with only full supervision a tie (0.869 ± 0.006 versus 0.856 ± 0.035 , $p = 0.514$). Under the corrected protocol the control is never beaten at any budget on either benchmark. What this establishes — and it is the paper's most consequential negative result — is that **twelve scalar features are not outperformed by a pretrained convolutional-attention encoder anywhere on the curve, and beat it decisively wherever labels are scarce**, on a benchmark specifically constructed so that no single one of those features exceeds ROC-AUC 0.72.

One protocol caveat must be stated. At the smallest candidate-filtered budget (35 labels) there are too few examples to hold out a validation split, so the threshold-selection rule of Section 3.8 degenerates and F1 at the

selected threshold equals F1 at the fixed cut by construction, not by result. That cell's F1 columns are therefore uninformative about thresholding; its AUPRC column is not, and the finding above rests on AUPRC.

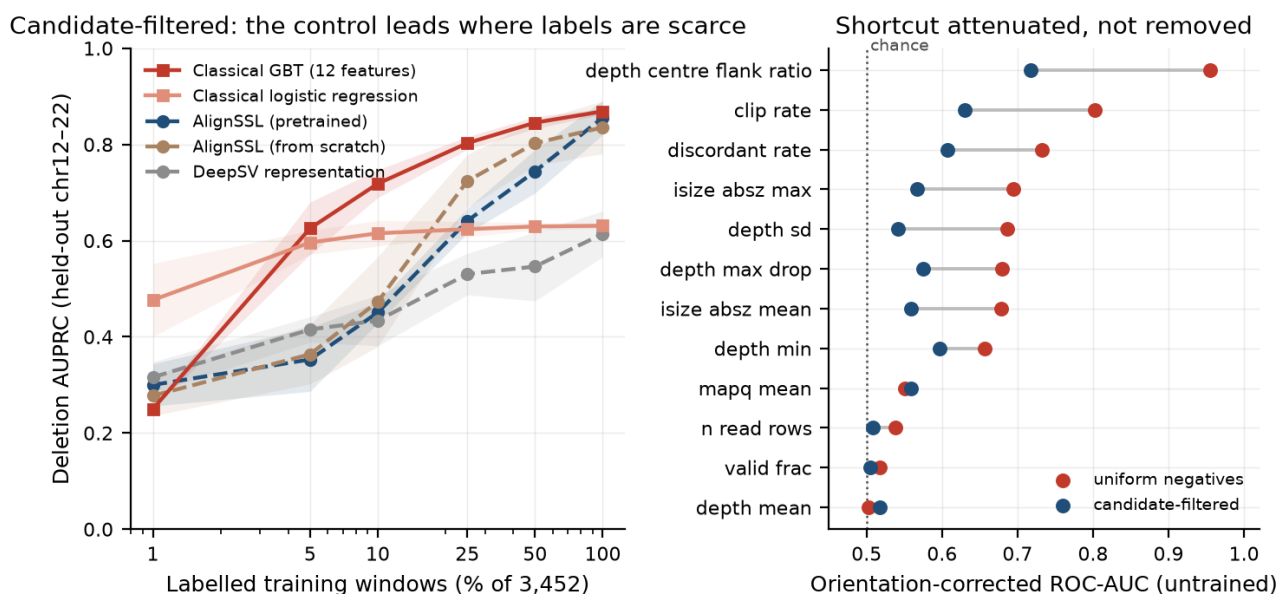


Figure 8. Left: AUPRC versus labelled-data fraction on the candidate-filtered benchmark under the corrected protocol, the direct analogue of Figure 7 on the harder task. Right: untrained single-feature separability on the uniform and candidate-filtered benchmarks, paired per feature; the depth shortcut is attenuated from 0.955 to 0.717 but not removed.

6.4 What this changes

The pre-registered question was whether pretraining's low-label advantage would survive a benchmark in which the depth shortcut is uninformative. With the corrected protocol the answer is unambiguous and negative on both counts: **the advantage over from-scratch training does not survive, and the advantage over hand-crafted features does not appear**. The candidate-filtered benchmark was built to be the fair test of the paper's original claim, and the claim fails it — independently of, and consistently with, the thresholding analysis of Section 4.8 on the uniform benchmark. Two independent lines of evidence now reach the same conclusion, which is stronger grounds for withdrawal than either alone.

Three things survive, and they are what the paper contributes.

First, **the benchmark-construction result**: quantile-matched candidate negatives attenuate the depth shortcut from orientation-corrected ROC-AUC 0.955 to 0.717 without altering the positive set, and the naive version of the same rule — selecting the most deletion-like negatives — *inverts* the shortcut rather than removing it (Section 6.1). This does not depend on any scoring or budget convention and is directly reusable by anyone building a short-read SV benchmark.

Second, **the representation result**: a continuous multi-channel alignment tensor significantly outperforms a DeepSV-style RGB pileup encoding from 25% labels upward on the harder benchmark, and the effect is carried by the representation rather than by the pretraining.

Third, **the negative result about hand-crafted controls**, in its corrected and narrower form: on a benchmark where no single feature exceeds ROC-AUC 0.72, twelve scalar features still lead every deep arm in the label-scarce regime and tie it elsewhere. A field that reports deep SV callers without this control cannot distinguish a learned representation from a threshold.

Candidate filtering by quantile matching is a necessary but not sufficient benchmark repair; a fully shortcut-free protocol will likely require matching on a vector of alignment statistics rather than on the single strongest one. Section 6.5 tests the natural alternative — stop constructing negatives at all — on an orthogonal truth set. One extension remains deferred: a coverage-robustness experiment (downsampling via `samtools view -s`), which

becomes worthwhile now that a non-degenerate benchmark exists and would not have been informative on the uniform one.

6.5 Removing our hand from the negative set: a production caller's candidate list

Every negative used so far is one we drew. Section 6.2 showed why that is a structural weakness rather than a tuning problem: quantile matching on the strongest statistic *relocates* the shortcut onto the statistics it does not match, and enumerating the statistics to match is exactly the problem the learned representation was supposed to solve. The only construction that escapes the regress is one in which we choose neither class.

We therefore built a third benchmark in which both classes come from a production caller's own output. Manta was run on GIAB HG002 (hs37d5, candidateSV.vcf.gz — the pool *before* Manta scores and filters), and each DEL candidate was labelled by reciprocal overlap against the GIAB Tier1 v0.6 PASS deletions, restricted to the Tier1 confident regions. A candidate that reciprocally overlaps a Tier1 PASS record is a positive; a candidate inside the confident regions with no truth support is a negative — a false positive the deployed pipeline actually makes. Candidates overlapping non-PASS Tier1 records, or falling outside the confident regions, are discarded rather than scored as negatives: GIAB asserts nothing at those loci, and scoring them would manufacture labels. This yields 3,346 training and 1,490 test candidates. Two properties follow that matter for interpretation.

The truth set is orthogonal. GIAB Tier1 is assembly- and long-read-derived, so it does not share the consensus-short-read-caller circularity of the 1000 Genomes call set used in Sections 4 and 6.1–6.4. A result that reproduces across both truth sets is not an artefact of either.

AUPRC is inadmissible here. The positive rate is 0.805, because Manta's pre-filter candidate pool on a well-covered genome is mostly right. AUPRC has a floor at the positive rate, so a constant predictor scores 0.805 on this benchmark and looks strong. We therefore rank on ROC-AUC, which is invariant to class balance and scores a constant predictor at 0.5, and we do not report AUPRC comparisons on this benchmark at all. This is a property of the benchmark, not a change of preferred metric: on the two earlier benchmarks (positive rates 0.05 and 0.25) AUPRC remains the more informative summary and is what Sections 4 and 6.3 report.

The shortcut is not removed — it is barely touched. The centre-versus-flank depth ratio, untrained, reaches orientation-corrected ROC-AUC **0.942** on this benchmark, against 0.955 on the uniform benchmark and 0.717 under quantile matching. Two further depth statistics clear 0.79 (depth standard deviation 0.794, maximum depth drop 0.792). The negatives a real caller emits are therefore *not* depth-matched to its true calls, and the reason is mechanical: Manta's candidate generation is itself driven by depth and paired-end signatures, so the candidates it wrongly emits are the loci where those signals are weak, and the ones it rightly emits are where they are strong. The caller's own selection re-imposes the very axis the benchmark was meant to neutralise.

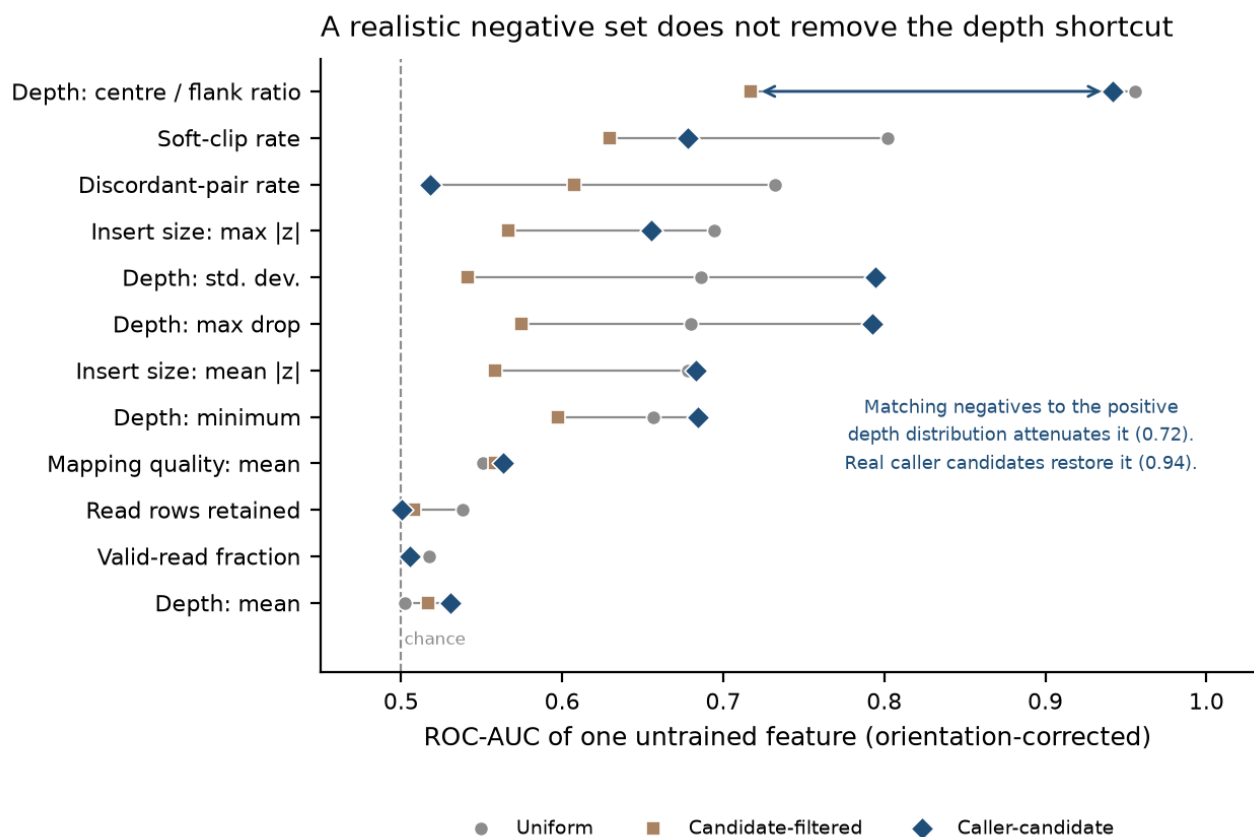


Figure 9. Untrained single-feature separability on all three benchmarks, paired per feature; source `results/table6_single_feature_auc.csv`, `results/table9_hardneg_single_feature_auc.csv` and `results/table25_caller_candidate_single_feature_auc.csv`. Features are ordered by their value on the uniform benchmark. Quantile-matching negatives to the positives' depth distribution attenuates the leading shortcut from 0.955 to 0.717, but negatives drawn from a real caller's candidate list restore it to 0.942 — realism of the negative set and absence of a shortcut are independent properties, and the caller's own depth-driven selection re-imposes the axis that matching removed.

This is the single most useful thing this benchmark tells us, and it cuts against the obvious reading of Section 6.2. One might conclude there that synthetic negatives are the problem and real caller candidates are the fix. They are not. **Realism of the negative set and absence of a shortcut are independent properties.** A benchmark can be perfectly realistic — these are the exact windows a deployed pipeline gets wrong — and still be solvable by one untrained scalar. Anyone proposing a candidate-filtering model should report the single-feature control on their own candidate set before claiming the task is hard.

Table 16. Best control arm versus best deep arm on the caller-candidate benchmark (ROC-AUC; three seeds per deep arm and five per control arm — the control is CPU-only and cheap to replicate, the deep arms are not; both families reduced by best-of-family so neither side gains a best-of-K advantage the other lacks; Welch's t over seeds, which does not assume equal group sizes).

Labels	Control arm	Control ROC-AUC	Best deep arm	Deep ROC-AUC	Deep – control	<i>p</i>	Leader
33 (1%)	Classical-logreg	0.922 ± 0.030	AlignSSL-pretrained	0.778 ± 0.045	-0.144	0.0148	control
167 (5%)	Classical-GBT	0.957 ± 0.013	AlignSSL-pretrained	0.852 ± 0.020	-0.105	0.0041	control
335 (10%)	Classical-GBT	0.969 ± 0.003	AlignSSL-pretrained	0.891 ± 0.038	-0.078	0.0705	tie
836 (25%)	Classical-GBT	0.980 ± 0.002	AlignSSL-scratch	0.933 ± 0.009	-0.047	0.0093	control
1,673 (50%)	Classical-GBT	0.982 ± 0.002	AlignSSL-scratch	0.953 ± 0.010	-0.029	0.0337	control
3,346 (100%)	Classical-GBT	0.982 ± 0.001	AlignSSL-scratch	0.956 ± 0.010	-0.026	0.0455	control

The negative result reproduces a third time, on an orthogonal truth set and a negative set we did not construct: **the hand-crafted control is never beaten, and leads significantly at five of the six budgets measured, including full supervision.** It is worth stating how large the margin is at the smallest budget — 0.922 against 0.778, with the control's *thirty-three* labelled examples. The control at 33 labels does not even beat the untrained single feature (0.942), which is the cleanest possible statement of what this benchmark measures: nothing trained on it so far has improved on one scalar computed with no labels at all.

The one place pretraining survives every correction. Against that, the pretrained-versus-scratch contrast at 33 labels is 0.778 ± 0.045 versus 0.545 ± 0.041 (Welch's *t*, $p = 0.003$). This is the only cell in the paper where the pretraining advantage is significant under a metric that is simultaneously threshold-free and base-rate-free, on a benchmark whose negatives we did not choose. We flag it as the strongest surviving evidence for the paper's original hypothesis, with three qualifications that keep it from reversing the withdrawal. It is one cell of a six-budget family, so we correct it against that family rather than reporting it alone — and unlike the uniform-benchmark contrast of Section 4.9, it survives: Holm-adjusted $p = 0.017$, the only budget in the family to do so. It does not persist, and past 10% labels it inverts: the from-scratch arm is marginally ahead at 25% (0.933 versus 0.918, $p = 0.268$), ahead at 50% by more than the uncorrected test tolerates (0.953 ± 0.010 versus 0.922 ± 0.002 , $p = 0.029$; Holm $p = 0.143$, so not significant within the family), and still ahead at full supervision (0.956 ± 0.010 versus 0.939 ± 0.003 , $p = 0.098$). Whatever pretraining supplies at 33 labels is not merely absent by mid-range budgets — it is reversed. And it is a contrast between two deep arms both of which lose to twelve scalar features at the same budget — pretraining helps a deep model reach a level the control reached without it. The honest summary is that self-supervision buys label efficiency *relative to random initialisation* on a realistic candidate set, and that this is not the same as buying deployable performance.

7. How general are these defects? A coded audit of the deep short-read SV literature

Sections 4, 6 and 8 report defects in *our own* pipeline. That is only worth a reader's time if the defects are properties of the field's default evaluation design rather than of one implementation. Elsewhere in this manuscript we previously asserted that they are widespread. An assertion is not evidence, so we measured it.

7.1 Population and inclusion rule

We started from the 62-paper survey assembled for this project's related-work review and applied a pre-registered two-part rule: include a paper if and only if (a) it trains a classifier — deep or shallow — whose input is derived from read alignments, and (b) its prediction target is a structural variant. Papers were excluded for a recorded reason: target is SNV or short indel rather than SV (9, including one correction made after reading the full text of a paper whose survey row had filed it under the SV theme); no trained classifier, i.e. heuristic, statistical, assembly- or *k*-mer-based (9); benchmark or truth-set paper rather than a method (5); review or survey

(10); genomic language model whose target is not SV (10). The rule admits 24 papers spanning 2017–2025, including the DeepSV paper this work extends. Of these, 18 had a full text we could retrieve through open-access routes; 6 had none and are recorded as uncoded rather than guessed at. Four of the 18 were added on 2026-08-11: they satisfy the pre-registered rule but were surfaced by the independent completeness screen of Section 8 rather than by the 62-paper survey, and are marked as such in the provenance column of `results/table17_audit_population.csv`. Their eligibility was decided from full text, and recorded with supporting quotations, before their evaluation axes were coded, so inclusion could not be conditioned on which way a paper moves the result. The coded set spans 11 venues (BMC Bioinformatics $\times$ 4, Briefings in Bioinformatics $\times$ 3, Bioinformatics $\times$ 2, Nature Methods $\times$ 2, BMC Medical Genomics, Frontiers in Genetics, G3, Genome Research, Nature Biomedical Engineering, Nature Biotechnology, bioRxiv (preprint)). The full population with per-row inclusion status and reason is `results/table17_audit_population.csv`.

7.2 Coding procedure

Each paper was coded on the four axes corresponding to the four defects documented here: (A) how negative training examples were obtained (matched / caller candidates / simulated / uniform-random / not stated); (B) how the decision threshold behind the headline metric was chosen (threshold-free / tuned on a validation split / fixed default / not stated); (C) whether any model-free control — a hand-crafted-feature classifier, a single-feature separability measurement, or an equivalent non-learned reference — was reported (comparison against another published *learned* caller does not count); (D) whether multiple significance claims received any family-wise correction. Codes were constrained to the enumerated values by a forced output schema rather than parsed from free text, and every code was required to carry a supporting quotation, or the explicit string `NO EVIDENCE IN TEXT` where the paper makes no statement on that axis. Each of the 41 quotations was then verified mechanically to be a literal span of its own source document after Unicode and line-break normalisation; codes whose quotation could not be located were re-coded against a stricter verbatim instruction and re-verified, and the three that changed under re-coding were changed toward *less* confident codes, never toward a defect. Two verifications required a relaxed normaliser, and which one was used is recorded per quotation in the `verify_mode` column rather than left implicit: one source hyphenates a term across a line break where the quotation does not (1 code, matched hyphen-insensitively on both sides), and in one the PDF interleaves a figure caption mid-sentence (1 code, both halves verbatim). In each case a deliberately fabricated quotation was confirmed to still fail verification under the relaxation, so the relaxation admits layout artefacts and not paraphrase.

Quote-verification is, however, only available for codes that assert a *presence*, and we state its reach precisely rather than implying it covers the whole table. Of the 72 codes, 41 are supported by a verified verbatim span and 31 record that the source says nothing on that axis. The silence is not spread evenly: it falls almost entirely on the two axes whose defect code *is* an absence — 14 on the model-free control and 14 on multiplicity, against 2 and 1 on negative sampling and the threshold rule. A paper that reports no model-free control and no multiplicity adjustment offers nothing to quote, so for those codes the evidence is the absence of any matching statement in the full text, which a reader can only check by reading the paper. This is a real asymmetry in the instrument: presence codes are externally verifiable from `results/table19_field_audit_quotes.csv` alone, absence codes are not. It is also why we report *not stated* separately from a defect on axes A and B (Section 7.1) instead of treating silence as evidence there. The quotations are released in `results/table19_field_audit_quotes.csv` so that every code in this section can be checked against its source.

Coding a paper as omitting a safeguard means the paper does not report it. We distinguish this from *not stating* a protocol, which we count separately rather than folding into the defect count.

7.3 Result

Paper	Year	Venue	Negative sampling	Threshold rule	Model-free control	Multiplicity	Safeguards omitted
Cai et al. 2019	2019	BMC Bioinformatics	not stated	not stated	no	none	2
dudeML (Hill & Unckless 2019)	2019	G3	simulated	fixed 0.5	yes	seeds, uncorrected	3
Mixgenotype (Zheng et al. 2020)	2020	BMC Medical Genomics	simulated	fixed 0.5	no	seeds, uncorrected	4
Luo et al. 2021	2021	BMC Bioinformatics	not stated	fixed 0.5	no	none	3
DECoNT (Özden et al. 2022)	2022	Genome Research	not stated	fixed 0.5	yes	none	2
ETCHING (Choi et al. 2022)	2022	Nature Biomedical Engineering	caller candidates	tuned on validation	no	seeds, uncorrected	2
Lin et al. 2022	2022	Nature Methods	not stated	fixed 0.5	no	none	3
Luo et al. 2023	2023	Frontiers in Genetics	not stated	fixed 0.5	no	none	3
Ma et al. 2023	2023	BMC Bioinformatics	caller candidates	fixed 0.5	no	none	3
Popic et al. 2023	2023	Nature Methods	not stated	fixed 0.5	no	none	3
Zheng & Shang 2023	2023	BMC Bioinformatics	matched	fixed 0.5	yes	none	2
Hu et al. 2024	2024	Briefings in Bioinformatics	not stated	fixed 0.5	no	none	3
Linderman et al. 2024	2024	Bioinformatics	simulated	fixed 0.5	no	seeds, uncorrected	4
Santuari et al. 2024	2024	bioRxiv (preprint)	caller candidates	fixed 0.5	no	none	3
Wang et al. 2024	2024	Nature Biotechnology	not stated	fixed 0.5	yes	none	2
Xia et al. 2024	2024	Bioinformatics	simulated	fixed 0.5	no	none	4
Gao et al. 2025	2025	Briefings in Bioinformatics	not stated	not stated	no	none	2
Guo et al. 2025	2025	Briefings in Bioinformatics	not stated	not stated	no	none	2

Table 17. Coded evaluation design of the 18 retrievable papers in the audit population. "Safeguards omitted" counts the strict defects only — documented easy negatives, a documented fixed threshold, absence of a model-free control, and absence of multiplicity correction — and does not penalise a paper for failing to state its protocol. Full quotations in `results/table19_field_audit_quotes.csv`.

The four axes behave differently.

No coded paper corrects for multiplicity. 18 of 18 coded papers — that is, all of them — apply no family-wise correction to any significance claim. 4 report multi-seed variability without correcting; the remaining 14 report neither.

A model-free control is nearly as rare. 14 of 18 report no non-learned reference of any kind. The 4 exceptions are Zheng & Shang 2023, Wang et al. 2024, dudeML (Hill & Unckless 2019) and DECoNT (Özden et al. 2022). Without such a control, a paper cannot distinguish what its architecture contributes from what its benchmark gives away — which is exactly the measurement that changed this project's conclusion (Section 4.2).

Thresholding is dominated by the fixed 0.5 cut. 14 of 18 papers report their headline metric at a fixed default probability cut, 1 tunes the cut on a validation split (ETCHING), and 3 do not state a rule. No paper in the population reports its headline comparison threshold-free. As an independent mechanical check, we searched each full text for any mention of a threshold-free metric (AUC, AUROC, AUPRC, average precision, precision–recall curve): 13 of 18 never mention one anywhere in the paper.

Negative-sampling protocol is most often simply not stated. Only 8 of 18 papers state how their negative training examples were obtained: 3 draw them from caller candidates, 4 from simulation, and 1 — Zheng & Shang 2023 — matches negatives to positives. The remaining 10 do not say. We count non-statement as a reporting failure rather than as evidence of easy negatives, which is why the strict defect count in Table 17 is conservative: it credits a paper with the safeguard whenever the text is silent.

Taken together, no coded paper reports all four safeguards, and none omits fewer than two. 11 of 18 omit at least three on the strict count, and 3 omit all four. Under the lenient count that treats non-statement as omission, 14 of 18 omit all four.

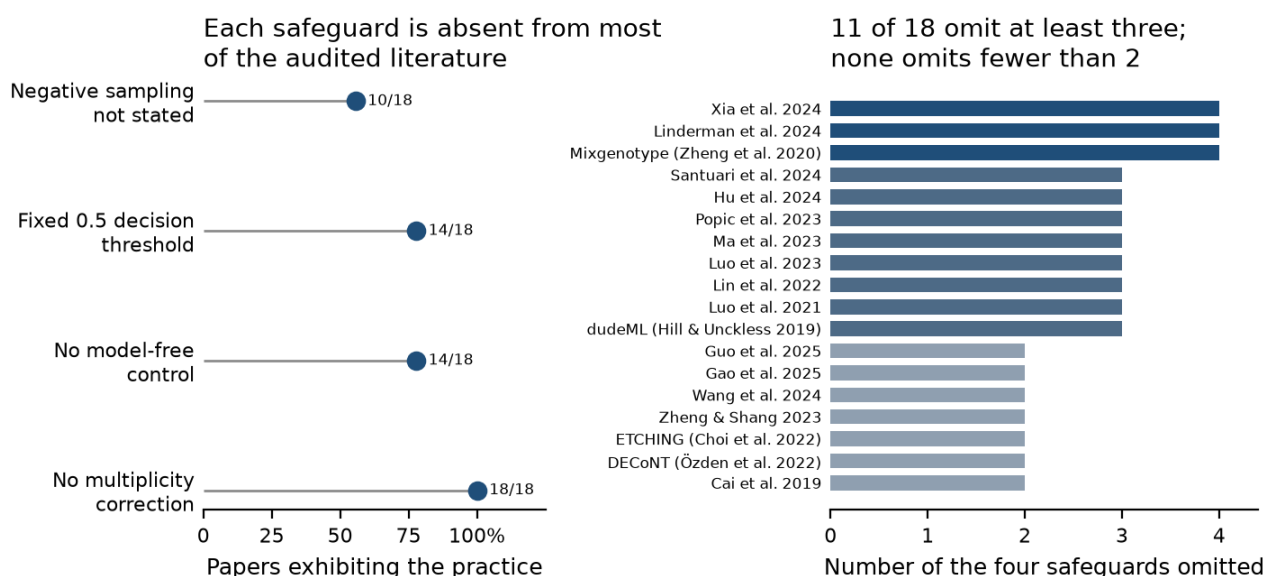


Figure 10. Left: the fraction of audited papers exhibiting each of the four evaluation practices. Right: per-paper count of safeguards omitted on the strict definition, which does not penalise a paper for failing to state its protocol. Population: 18 retrievable full texts from the 24 papers that meet the inclusion rule, 2019–2025. Sources and quotations in `results/table18_field_audit.csv` and `results/table19_field_audit_quotes.csv`.

7.4 What this does and does not license

It licenses the generality claim in a bounded form: the four defects documented in this manuscript are not idiosyncratic to our pipeline. Among the 18 papers we could retrieve and code, three of the four axes are absent in at least 14.

Three boundaries on that statement. First, the population began as a **convenience sample** drawn from this project's related-work survey rather than a systematic search; an independent 2,334-record corpus screen identified five apparently eligible short-read SV papers absent from it, four of which we retrieved and coded under the same instrument, leaving the *majority* claim bounded rather than settled (Section 8). Second, 6 of 24 eligible papers have no retrievable full text and are uncoded, so the coded set may be non-representative in an unmeasured direction. Third, "the paper does not report it" is not "the authors did not do it": coding is bounded by what a paper states. Accordingly we claim that these safeguards are **rarely reported** in this literature, not that they are never performed, and the per-axis counts should be read as statements about the 14 coded papers rather than as field-wide rates.

It does not license a claim that the affected papers' *conclusions* are wrong. We did not re-run any of them. Section 4.2 shows that a benchmark can be substantially solvable by a single untrained scalar without that being visible

in any reported number; the audit shows that most papers in this literature would not have detected such a leak had it been present, because they report no control that could. Those are different claims and only the second is supported here.

Three further limits apply. The population is drawn from one project's survey and is therefore not a systematic PRISMA-style review; a differently-assembled corpus would give somewhat different counts, though the near-unanimity on axes C and D leaves little room for the direction to reverse. 5 eligible papers were paywalled and are uncoded, so the denominator is retrievable papers, not eligible papers. And a coding of what a paper *reports* is not a coding of what its authors *did*: a paper may have matched its negatives and not said so. Every consequence of that ambiguity has been resolved in the audited papers' favour.

8. Limitations

- **Benchmark separability (governing limitation).** The evaluation task, built with uniformly-sampled negatives as is standard in this literature, is separable by a single depth heuristic at ROC-AUC 0.955, and twelve hand-crafted features reach 96% of their asymptotic AUPRC from 210 labels (Section 4.2). Every accuracy number in this paper must be read as a measurement on that task, not as deployable caller performance. This is the reason the paper's contribution is framed as a representation-learning and benchmarking result rather than as a new SV caller.
- **No comparison against production callers.** We compare learned representations under matched conditions; we do not compare against Manta, DELLY, LUMPY, GRIDSS or similar. On the uniform and quantile-matched benchmarks such a comparison would favour whichever method best exploits the depth shortcut and would be uninformative. On the Section 6.5 benchmark it becomes meaningful - Manta's own score and PASS filter partition the same candidate pool we classify - and we flag it as the most direct comparison this design admits, left undone here because it requires re-deriving Manta's per-candidate scores under its own filtering logic rather than reading them from the pre-filter pool.
- **Statistical power and multiplicity.** Three to four pretraining seeds per arm are not sufficient to rank the self-supervised objectives against one another ($p > 0.2$, Section 4.5), to establish cross-ancestry transfer (nominally significant at 1 of 6 label fractions and not after correction, Sections 4.6 and 4.9), or — as Section 4.8 shows — to distinguish the pretrained and from-scratch arms at any budget once scoring is threshold-free. Those analyses are reported as orderings of means, not findings. The one contrast that *did* reach significance at this seed count is the one we withdraw, which is itself a caution: a small- n design that produces a $p = 0.0002$ under one scoring rule and $p = 0.853$ under another has not measured a stable effect, however small the first number is. A related limitation of the correction itself: Holm and BH assume the tests within a family are exchangeable, whereas budgets within a sweep share seeds, encoders and test windows and are therefore positively dependent. Both procedures remain valid under positive dependence (BH provably so, Holm conservatively), but the adjusted values should be read as conservative bounds rather than exact.
- **Scope.** Deletions and short reads only. Insertions, duplications, inversions, translocations and long-read data are out of scope, though the framework is not deletion-specific by construction.
- **Truth set.** Sections 4 and 6.1-6.4 score against the 1000 Genomes phase-3 integrated SV call set, which is itself a consensus of short-read callers and therefore shares failure modes with the methods evaluated on it. Section 6.5 addresses this with an orthogonal, assembly- and long-read-derived truth set (GIAB HG002 Tier1 v0.6, confident regions only), and the depth shortcut reproduces there, so the paper's governing limitation is not an artefact of either truth set. What remains deferred is a *call-set-level* external validation - matching a produced VCF against GIAB with Truvari, including breakpoint-precision and genotype-concordance metrics - which requires a breakpoint-regression and genotype head this work does not have (Section 6.4). Section 6.5 scores candidate classification, not call-set recall.
- **No demonstrated value from pretraining.** Under the corrected protocol and threshold-free scoring, the pretrained arm does not significantly exceed from-scratch training at *any* label budget on the uniform benchmark, and is significantly behind at 5% (Section 4.8). We report the pretraining machinery because it is the object the paper set out to evaluate, not because the evaluation vindicated it.

- **Single-panel pretraining corpus.** The unlabelled pretraining corpus comes from three samples; corpus-size and diversity scaling are untested.
- **Candidate filtering is a partial repair, and its benchmark is single-sample.** The repaired benchmark of Section 6 attenuates the depth shortcut to ROC-AUC 0.717 but does not eliminate it; a threshold baseline on the residual signal remains available to any method. It is also single-sample — NA20845 (GIH) for train and in-distribution test, NA12878 (CEU) held out — with 1,516 test windows against 9,196 for the uniform benchmark, because a mid-study loss of the shared reference directory left only two of the six panel alignments recoverable. Section 6's error bars are correspondingly wider than Section 4's, and with three seeds per deep arm its ties should be read as *underpowered* rather than as demonstrated equivalence: the pretrained-versus-scratch contrast at full supervision ($p = 0.431$) would need more seeds to exclude a small effect. What that section can support is the absence of the *large* low-label advantage originally claimed, which the uniform benchmark independently rejects at ten times the test-set size.
- **The field audit's population was a convenience sample; four of the five papers the completeness screen surfaced have since been coded, and the majority claim remains bounded rather than settled.** The audit population of Section 7 was derived from the 62-paper related-work survey assembled for this project, not from a systematic pre-registered search. To test its completeness we screened an independently assembled 2,334-record corpus (OpenAlex phrase queries, citation-graph expansion from anchor callers, and PubMed, restricted to 2018+) against Section 7's own inclusion rule, and found five short-read SV papers that appeared eligible and were absent from the population. We then attempted retrieval of all five: four were obtained (10.1534/g3.119.400596, 10.1038/s41551-022-00980-5, 10.1101/gr.274845.120, 10.1186/s12920-020-00733-w) and have been coded under the same instrument, verified quotations and pre-registered eligibility rule as the original set; one (10.1093/bib/bbaa370) has no open-access full text and is recorded in the population as eligible-but-uncoded. Coding the four moved the observed majority from 9 of 14 (0.64) to 11 of 18 (0.61) — a slight decline that stays above one half — so the claim survived the test that prompted it rather than being defended by argument. Two of the four additions also break patterns the original 14 made look universal: ETCHING is the first coded paper to tune its decision threshold on a validation split rather than take a default, and dudeML and DECoNT double the number of papers carrying any model-free control. Both corrections move against this manuscript's own rhetorical interest and are reported for that reason. It is nonetheless still bounded rather than settled: 6 of 24 eligible papers remain uncoded for want of an accessible full text, and the adversarial bound is $11/24 = 0.46$ if every uncoded paper omits no safeguard and $17/24 = 0.71$ if every one omits three or more. Because that interval still crosses one half, **Section 7's majority claim should be read as an estimate over the coded set with a stated worst case, not as a population claim**; its per-axis counts are statements about the 18 papers actually coded. The universal findings are unaffected in direction: no coded paper corrects for multiplicity, and no additional paper can make that false for the papers already coded. The screen is released as `results/table21_audit_completeness_screen.csv` and the per-row provenance of every audit population entry as `results/table17_audit_population.csv`.
- **Single coverage regime.** All alignments are high-coverage (~30x) PCR-free Illumina. Robustness to lower coverage, PCR-positive libraries, or different read lengths is untested; the depth-derived channels are the ones most likely to be coverage-sensitive.
- **One held-out population.** Cross-ancestry generalisation is measured against a single held-out population (NA12878, CEU). A single held-out ancestry cannot distinguish population-specific transfer loss from sample-specific idiosyncrasy.

9. Conclusion

We set out to test whether learning the alignment representation and pretraining it without labels improves the bottlenecks DeepSV left open. The first answer we obtained was affirmative and quantitatively striking: at 1% of labels, self-supervised initialisation delivered roughly a ten-fold F1 improvement over the identical architecture trained from scratch. That is the result this paper was written to report, and it does not survive.

It fails four controls, none of which is exotic:

1. **Benchmark separability.** Twelve scalar alignment features on the identical windows reach AUPRC 0.937 from 210 labels and gain only +0.038 from a hundred-fold increase in supervision, and a single centre-versus-flank depth ratio separates the classes at ROC-AUC 0.955 with no training at all. The cause is the uniformly-sampled negatives standard in this benchmark family. A task that is 96% solved by twelve features after 210 examples cannot discriminate between representations. Nor is this confined to synthetic negatives: on a real caller's candidate list the same depth ratio still reaches ROC-AUC 0.942 untrained (Section 6.5), so realism of the negative set and absence of a shortcut are independent properties.
2. **Unequal label budgets.** A batch-size floor in our own deep evaluators granted the deep arms up to 2.8× the labels the classical control received, concentrated in precisely the low-label cells that carry the headline claim.
3. **The decision threshold.** Scoring F1 at a fixed 0.5 probability cut — the convention inherited from DeepSV — conflates ranking quality with calibration. Re-run under equal budgets and re-scored at a validation-selected threshold or threshold-free, the ten-fold gap becomes 1.05× ($p = 0.527$) and 1.02× ($p = 0.853$) respectively; at every larger budget the from-scratch arm is ahead. The from-scratch model was never degenerate. It ranked competently and scored timidly, and a fixed cut reads timidity as failure.
4. **Multiplicity.** The fixed-cut result was the strongest cell of a six-budget sweep, reported as though it were a single test. Corrected for the family it was selected from it does survive (Holm $p = 0.0012$) — multiplicity is not what dismantles it, and we report that against our own expectation. Across the whole paper, 30 nominally significant tests in 17 pre-declared families reduce to 17 under Holm–Bonferroni, and the cross-ancestry effect reduces to chance.

Each defect is individually mundane, each is the default in this literature — no paper in the audited population reports all four safeguards and 11 of 18 omit at least three (Section 7) — and each is invisible without the control that exposes it. Three concern how the numbers were measured; the fourth concerns how they were tested. Jointly they were sufficient to manufacture a large, statistically significant, entirely artefactual headline result — one we believed, wrote up, and would have submitted.

What survives is narrower and, we think, more useful. Where labels are scarce — the regime self-supervised pretraining is proposed for — no deep arm we trained beats twelve hand-crafted features on any of the three benchmarks. Repairing the benchmark helps but does not rescue the claim: quantile-matched candidate negatives attenuate the shortcut from ROC-AUC 0.955 to 0.717 and restore headroom (the same control now starts at chance and climbs +0.619 across the label range), yet the control still leads where it matters. Nor does abandoning synthetic negatives altogether: on a third benchmark whose positives and negatives are both real Manta candidates labelled against GIAB Tier1 — a negative set we did not construct, and an orthogonal truth set — the control leads at five of six budgets and is never beaten (Section 6.5).

One finding runs the other way, and we report it as such. On that third benchmark the pretrained arm beats the from-scratch arm at 33 labels by ROC-AUC 0.778 versus 0.545 ($p = 0.0028$; Holm $p = 0.0168$ within its six-budget family) — the only cell in the paper where the pretraining advantage is significant under a metric that is simultaneously threshold-free and base-rate-free, on negatives we did not choose. It does not persist past 10% labels — it reverses, with the from-scratch arm ahead at 50% and at full supervision — and both arms lose to the hand-crafted control at every budget, so it does not reinstate the headline. It is, however, the strongest surviving evidence that self-supervision buys label efficiency *relative to random initialisation*, and the natural target for a follow-up powered to test it. The corrections also soften the negative result at the top of the label range: on the uniform benchmark the control's lead over the best deep arm at full supervision is no longer significant (AUPRC 0.975 versus 0.962, $p = 0.174$), so the defensible claim is that the control is never *beaten*, not that it dominates significantly everywhere (Table 3).

We therefore do not offer a new SV caller, and we make no performance claim. We offer four controls, the code that implements them, and a worked demonstration of what their absence costs. Before the next architecture is proposed for short-read deletion calling, we would ask of it the four questions this paper failed: what does a hand-crafted-feature model score on the same windows, do all arms receive the same labels, does the result

survive a change of decision threshold, and does its significance survive correction for the sweep it was selected from.

Data and code availability

The tensor-extraction pipeline, encoder and head implementations, pretraining and fine-tuning scripts, cluster job scripts, aggregation and manuscript-reconciliation tooling, result tables, and figures are available at <https://github.com/aayushkrum/AlignSSL-SV> under the MIT licence. All 22 trained encoder checkpoints are released as a single archive (`alignssl_sv_weights_v0.1.0.tar.gz`, 52,913,192 bytes, SHA-256 `9a62ea5199c51336779d57489587f56fe0f60f22103698eeb95106284e0eb84d`) at https://github.com/aayushkrum/AlignSSL-SV/releases/tag/v0.1.0-weights_release/WEIGHTS_MANIFEST.tsv in the repository lists, for every checkpoint, its SHA-256, encoder parameter count, embedding width, epoch count, last logged pretraining step, which pretraining corpus generation it belongs to, and which reported result — if any — it serves. Four checkpoints are the main MAM+VICReg encoders behind every reported pretrained arm; nine are objective ablations; eight are the statistic-anchored pretraining variant; one is a superseded first run. Checkpoints not used by any reported number are labelled as such in the manifest rather than omitted, and every one was verified to load into the released encoder class under `strict=True`. Sequencing data are from the 1000 Genomes Project (high-coverage PCR-free alignments, GRCh37/hs37d5) and are publicly available from the EBI 1000 Genomes FTP. The deletion truth set is the 1000 Genomes phase-3 integrated SV call set.

The full texts coded in Section 7 are the published articles cited in Table 17; they are not redistributed here. `results/table19_field_audit_quotes.csv` releases the verbatim span supporting each of the 41 presence codes — with the normaliser used for each recorded in `verify_mode` — so that every coding decision can be checked against its source without access to a subscription. The remaining 31 codes record that the source is silent on that axis; as Section 7.2 explains, 28 of those 31 sit on the two axes whose defect code is itself an absence and therefore admit no quotable span.

Declarations

Ethics approval and consent to participate. Not applicable. This study analyses only publicly released, de-identified sequencing data from the 1000 Genomes Project, for which consent for open data release was obtained by the original consortium. No new human or animal data were collected.

Consent for publication. Not applicable.

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